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Optimizing stock market prediction and stock trading strategies with deep learning models enhanced by nonlinear feature identification and robust prediction evaluation

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Abstract

This study investigates cross-sectional stock return prediction in the A-share market from 2013 to 2024 with equity and firm-characteristic data available at databases such as RESSET and CSMAR for more than 5,000 listed firms. We introduce the Diff-RMSE method for nonlinear factor identification and the weighted evaluation index (WEI), a finance-specific performance metric that integrates prediction accuracy with market adaptability. Based on 30 market, liquidity, valuation, profitability, technical and risk factors, we compare linear models, tree-based machine learning and deep learning architectures—including GRU, LSTM and Transformer—within a rolling-window forecasting framework. We further translate return forecasts into long–short portfolios to assess economic performance. Our results show that deep learning models, particularly LSTM and Transformer, deliver superior accuracy, more stable WEI scores and stronger tail-risk control than traditional benchmarks. These findings provide practical guidance for quantitative portfolio managers and enrich the literature on machine-learning-based stock-selection models in stock markets.

Keywords: Deep learning, Stock return prediction, Transformer, Weighted evaluation index, Nonlinear feature identification, Stock trading strategy

Introduction

Over the past decade, the landscape of empirical asset pricing has shifted from traditional linear factor models toward increasingly flexible approaches capable of capturing nonlinear and unstable return-generating processes. While multifactor structures to more comprehensive multifactor specifications capturing size, value, momentum, profitability, and other anomalies (Carhart 1997; Fama & French 2015; Hou et al. 2015) remain influential, their linear and static specifications are often inadequate in the presence of evolving market microstructure, state-dependent factor premia, and pervasive interactions documented in recent "factor-zoo" research (Feng et al. 2020). These limitations have motivated the adoption of machine learning (ML) methods to model high-dimensional, nonlinear, and time-varying patterns more effectively (Chen et al. 2024; Gu et al. 2020; Jiang et al. 2023).

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Recent ML progress has expanded the methodological toolkit available to asset pricing researchers. Tree-based models such as gradient boosting machines capture threshold effects and complex variable interactions without restrictive parametric assumptions (Bryzgalova et al. 2019; Ślusarczyk & Ślepaczuk 2025). Deep learning architectures—including LSTM networks and transformer-based models—further enhance representational capacity, allowing the extraction of latent structures embedded in financial characteristics (Hochreiter & Schmidhuber 1997; Vaswani et al. 2017). However, these gains in predictive flexibility come with challenges: deep architectures often lack transparency, posing difficulties for interpretation, risk management, and institutional adoption (Molnar 2020; Mullainathan & Spiess 2017). As a result, interpretability and evaluation frameworks have become central concerns in modern quantitative finance. In parallel, the growing emphasis on robustness has elevated the importance of rolling-window forecasting, which mimics real-time model deployment, mitigates look-ahead bias, and evaluates adaptability under shifting market regimes (Aiolfi et al. 2010; Welch & Goyal 2008). Techniques such as Shapley value analysis provide essential insights into model behavior by decomposing predictions and identifying economically meaningful drivers—even for sophisticated deep models (Goulet Coulombe et al. 2022; Lundberg 2017; Shapley 1953). Together, these developments underscore an urgent need for frameworks that integrate (i) flexible modeling, (ii) interpretable factor structures, and (iii) finance-specific evaluation metrics capable of capturing both predictive accuracy and economic relevance.

This study addresses this gap in three ways. First, to disentangle nonlinear factor importance, we employ Diff-RMSE, a model-agnostic recursive feature elimination approach that quantifies each factor's marginal predictive contribution across rolling windows. Second, we propose a new performance metric—the weighted error index (WEI)—that embeds cross-sectional asymmetry and directional accuracy into model evaluation, complementing conventional loss metrics and extending mean adjusted directional loss (MADL) and generalized MADL (GMADL). Third, leveraging insights from Diff-RMSE and attention-based architectures, we construct deep-learning-driven long–short portfolios for the A-share market, demonstrating that models such as LSTM and Transformer deliver more stable forecasts, superior long–short returns, and enhanced tail-risk control relative to linear and tree-based benchmarks. Taken together, the contributions of this paper are threefold:

(1) Methodological contribution—nonlinear factor identification.

We introduce Diff-RMSE to quantify the marginal and interaction-based contributions of 30 predictive factors across rolling windows, providing a principled approach to nonlinear factor screening in high-dimensional settings.

(2) Evaluation contribution—finance-specific performance metric (WEI).

We develop the weighted error index (WEI), which incorporates both error magnitude and directional frequency, capturing cross-sectional imbalance and market asymmetry in a manner tailored to stock-selection contexts.

(3) Empirical and practical contribution—deep learning-driven strategies for trading of equities.

Using a twelve-year sample of A-share stocks, we show that deep architectures—especially GRU and Transformer—yield materially stronger long–short performance and lower tail risk than traditional models, offering actionable guidance for quantitative portfolio managers operating in emerging markets.

Literature review

Factor-based asset pricing models

Research on factor-based investing traces back to the CAPM first proposed by Lintner (1965). The CAPM initially posited that the market beta serves as the sole systematic determinant of returns. However, early empirical inquiries uncovered several anomalies, calling into question whether a single factor could adequately account for cross-sectional differences in stock returns. In response, Fama and French introduced size and value as additional factors, demonstrating that these metrics account for a significant fraction of observed return variation. Shortly thereafter, Carhart (1997) proposed a momentum factor, reflecting that stocks with recent gains tend to outperform in the near term. Later refinements built on this framework, including the five-factor model of Fama and French (2015), which incorporates profitability and investment factors, as well as the q-factor model of Hou et al. (2015). Nonetheless, these models still rely on linear specifications and static loadings, potentially limiting their ability to represent time-varying or nonlinear relationships.

More recent efforts have drawn on high-frequency data to develop new factors with strong predictive performance in specific market conditions. Gu et al. (2020) demonstrate that machine learning can uncover new predictive factors, thereby extending the traditional linear framework. By examining numerous potential market anomalies, their approach dynamically adapts to time-varying return relationships and surpasses conventional factor models in both in-sample estimation and out-of-sample testing. Similarly, Feng et al. (2020) propose a strict factor screening procedure that addresses the "factor zoo" phenomenon and systematically evaluates newly introduced factors. Their method seeks to avoid overfitting when researchers explore a large pool of candidate factors, ensuring that spurious or coincidental predictors are excluded. Hou et al. (2018) likewise replicate over four hundred published market anomalies and find that most are not robust under stricter empirical scrutiny. These studies underscore the importance of thorough factor verification.

Researchers are now using alternative data sources—such as corporate textual disclosures, social media, and consumer transaction records—to create more context-sensitive and timely factors. Jiang et al. (2019) show that analyzing earnings call transcripts can reveal sentiment-driven factors linked to future returns. Corporate filings and social media content are also being used to create specialized predictors (Wang et al. 2024). By incorporating sentiment from corporate announcements and social media, researchers can better account for institutional frictions and unique behavioral traits in China's market (Jabeen et al. 2022). These unstructured datasets can capture short-term risk premia or behavioral biases that traditional factors (e.g., size, value, momentum) may

miss. However, the reliance on high-dimensional inputs and intricate algorithms presents challenges for interpretability, stability, and implementation.

Machine learning in finance

From linear to nonlinear models

Linear models are popular for their simplicity and interpretability but often struggle with complex variable interactions. For example, Rapach et al. (2013) used Lasso to forecast global stock returns, incorporating lagged returns from multiple countries. Lasso highlights the most predictive features through L1 penalty implementation (Chun et al. 2022). While traditional linear regression and regularization-based models remain prevalent, researchers have increasingly examined model performance in nonlinear, structural breaks, and high-dimensional scenarios (Chen et al. 2024; Gu et al. 2021). Gu et al. (2020) used various machine learning methods (including tree models and deep neural networks) to systematically compare US stock cross-sectional returns and found that nonlinear models excel at capturing factor interactions and time-varying risk premiums while maintaining strong scalability and prediction accuracy in ultrahigh-dimensional feature spaces. Feng et al. (2020) focus on the "factor jungle" problem, cautioning against false factors caused by data mining in high-dimensional data environments through rigorous testing and selection of numerous candidate factors. However, these methods, despite performing well in out-of-sample prediction and trading strategy returns, also pose challenges to model interpretability and robustness (Mullainathan & Spiess 2017). Practical investment scenarios may face difficulties in regulatory compliance, risk assessment, and investor communication due to overly complex model structures. Therefore, striking a balance between flexibility and interpretability has become an important research direction (Giglio et al. 2022).

Deep learning architectures

Early financial research demonstrated the feasibility of neural networks in forecasting derivative pricing, particularly options (Hutchinson et al. 1994; Yao et al. 2000). Compared to traditional machine learning, deep neural networks overcome the limitations of simpler models by excavating potential nonlinear relationships in return data through multiple layers of representation (Chen et al. 2024; Heaton et al. 2017; Ślusarczyk & Ślepaczuk 2025). Researchers have designed various network architectures to address different data characteristics, enhancing model adaptability and predictive accuracy. For instance, multilayer feedforward networks can approximate any measurable function and exhibit higher flexibility in dealing with complex interactions of asset pricing factors (Gu et al. 2020). However, its "black box" characteristics also pose challenges to factor interpretation. Recent research aims to enhance model interpretability by integrating attention mechanisms and visualization techniques, making it easier for investors and regulators to understand internal operational mechanisms.

Convolutional neural networks (CNNs) can effectively capture local patterns and translation invariant features when processing financial data through "visualization" or matrix processing (such as price paths, technical indicator maps, etc.). Jiang et al. (2023) used CNN to extract patterns in stock price images and found significant differences between these patterns and common trend signals such as momentum or reversal. Their

method performed well in predicting returns and trading strategy returns and remained stable under various model settings. For high-frequency trading data, macroeconomic time-series data, and event-driven company data, long short-term memory networks (LSTMs) have significant advantages in capturing long-term memory and temporal dependencies (Fischer & Krauss 2018).

A more recent line of research introduces attention mechanisms and transformer architectures into financial time-series modeling, aiming to overcome the limitations of recurrent models in handling long sequences and capturing multiscale dependencies. Ding et al. (2020) propose a hierarchical multiscale Gaussian transformer to predict stock movements by jointly modeling fine-grained and coarse-grained temporal patterns, showing that attention can adaptively focus on periods with stronger information signals. Zhou et al. (2021) further advance the literature with *Informer*, an efficient Transformer variant that employs ProbSparse attention to reduce quadratic complexity and enhance long-horizon forecasting performance. These innovations have inspired a growing body of applications in financial forecasting, volatility prediction, and cross-asset return modeling, where long-range dependencies, structural breaks, and heterogeneous factor dynamics are critical. Compared with traditional LSTM or CNN architectures, transformers offer more flexible temporal receptive fields and better scalability to high-dimensional features. Building on these insights, our study incorporates transformer-based designs to capture cross-sectional nonlinearities and temporal attention patterns while adapting them to the unique characteristics of firm-level factor data.

Shapley analysis and model interpretability

The Shapley value analysis, originating from cooperative game theory (Shapley 1953), is a prevalent approach for elucidating the internal mechanisms of "black box" models. It evaluates variable contributions to model outputs by measuring incremental effects across all possible feature subsets (Lundberg 2017). The method allocates "credit" based on marginal contributions, establishing it as a relatively equitable tool for feature attribution. Recent financial empirical and econometric studies have employed Shapley values to analyze variable weights in complex algorithms (Goulet Coulombe et al. 2022). This approach is particularly valuable in multifactor environments for distinguishing variable influences. However, computational demands increase exponentially with variable scale, necessitating approximation or sampling techniques. Attribution results may become ambiguous when factors exhibit high overlap, such as between valuation indicators such as P/E and P/B (Nezlobin et al. 2016). Nevertheless, Shapley values remain instrumental in dissecting machine learning model operations within finance. Their quantitative decomposition of feature importance provides insights beyond traditional model diagnostics.

While Shapley values serve as a benchmark for feature attribution, researchers have explored alternative approaches to achieve breakthroughs in computational efficiency and local interpretability. LIME (Local Interpretable Model-Agnostic Explanations), for instance, explains model output variations in response to minor input space changes near individual observations through simplified local linear approximations (Ribeiro et al. 2016). However, LIME only elucidates the local influence of certain predictors in specific instances, lacking the capability to provide comprehensive factor rankings

across entire datasets. In contrast, counterfactual explanations identify minimal adjustments to data point attributes that yield different predictive outcomes (Wachter et al. 2017). Zhong and Enke (2019) employed various machine learning and deep learning algorithms to predict daily stock return directions while emphasizing the significance of interpretability techniques in empirical analysis. The study demonstrates that machine learning methods can efficiently optimize trading strategy parameters, achieving near-optimal performance with significantly reduced computation time compared to exhaustive search (Ryś & Ślepaczuk 2019). Grudniewicz and Ślepaczuk (2023) finds that machine learning-based trading strategies, particularly linear SVM and Bayesian GLM, outperform passive benchmarks across major indices in terms of risk-adjusted returns.

Rolling-window forecasting

The use of a single, fixed estimation window is often inadequate for capturing crucial changes in underlying processes. In equity and fixed income markets, macroeconomic adjustments or policy interventions can suddenly reshape return dynamics. Rolling window frameworks align more closely with real-time decision scenarios by training models using only recent historical data before making subsequent predictions (Welch & Goyal 2008). Through continuous forward movement of estimation windows, researchers can evaluate how models absorb new market data without relying on future information, enabling more rigorous examination of out-of-sample predictive performance while mitigating overfitting risks (Aiolfi et al. 2010).

This approach demonstrates advantages in market predictability studies. When structural breaks occur due to investor sentiment, macroeconomic fluctuations, or regulatory reforms, model coefficient stability may be compromised by overly long or static estimation intervals (Rapach et al. 2013). Rolling window predictions typically provide more timely parameter estimates by discarding obsolete observations and incorporating recent market signals during each re-estimation. This characteristic is especially valuable for practitioners who regularly update return or volatility forecasts, as it enables models to adapt more flexibly to evolving market conditions.

The selection of rolling window length necessitates balancing multiple considerations. Longer windows provide richer data samples, enhancing precision in relatively stable market environments (Zhang & Skiena 2010). However, when earlier observations no longer reflect current conditions, dated data may diminish the recognition of recent policy changes or structural break signals. Conversely, shorter windows, while more attentive to recent data, increase estimation variance—an effect particularly pronounced for models requiring substantial samples, such as deep neural networks (Tavakoli et al. 2023). This trade-off becomes especially evident in equity markets, where sudden reforms, listing mechanism changes, or policy adjustments rapidly alter market liquidity and investor behavior (Liu et al. 2019). While timely adjustments to rolling or expanding windows enhance opportunities to capture return patterns in specific market environments, they must ensure parameter estimation stability and avoid issues from frequent recalibration. Consequently, both researchers and portfolio managers must balance reflecting market evolution with obtaining reliable parameter estimates—a balance typically determined through empirical testing to identify the window size most suitable for their environment.

Evaluation metrics and synthesis

Building on recent advances in factor-based asset pricing and model interpretability, this study introduces a set of innovative evaluation metrics and risk management techniques designed to strengthen both the robustness and practical applicability of financial prediction models. Conventional error measures such as RMSE and MAE are useful for summarizing average forecast deviations, but they offer limited insight into tail risks, market regime shifts, and the directional quality of predictions—dimensions that are critical for real-world trading and risk control.

To address these gaps, we incorporate MADL and GMADL, which explicitly emphasize directional accuracy and asymmetric forecast penalties. Unlike symmetric loss functions, MADL directly penalizes incorrect return sign predictions, reflecting the well-established importance of direction forecasting for trading profitability and risk management (Christoffersen & Diebold 2006). he GMADL further magnifies losses associated with large adverse movements, making it particularly suited for environments where downside risks are disproportionately costly.

In addition, we propose two complementary innovations: the weighted evaluation index (WEI) and the Diff-RMSE metric. The WEI integrates predictive accuracy with sensitivity to changing market conditions, providing a more holistic and context-aware assessment of model performance than traditional point-error metrics. Meanwhile, the Diff-RMSE framework evaluates the marginal contribution of individual factors—and their combinations—to predictive accuracy, thereby complementing Shapley-based interpretability. This approach not only supports more effective factor selection but also yields deeper insights into interaction effects and the structural drivers of model performance. Together, these advances overcome the limitations of conventional evaluation metrics, enabling a richer assessment of model accuracy, robustness, and economic relevance in dynamic financial markets.

Methodology

Our methodological framework follows a streamlined pipeline for evaluating factor-based return prediction in an out-of-sample setting. After preprocessing the cross-sectional features, we implement a rolling-window procedure in which models are trained on historical data and used to forecast next-period returns, thereby mimicking real-time decision-making while avoiding look-ahead bias. We compare multiple model classes—linear, tree-based, and deep learning—to capture different degrees of nonlinearity. Feature importance is assessed using SHAP and Differential-RMSE, which quantify marginal contributions and interaction effects among predictors. Model forecasts are evaluated using both conventional error measures and finance-specific metrics that account for directional accuracy and asymmetry. Finally, predicted returns are used to form long–short portfolios by ranking stocks from top to bottom. Portfolio performance is assessed through risk-adjusted statistics. This integrated framework enables a rigorous comparison of forecasting methods and their economic implications (Fig. 1).

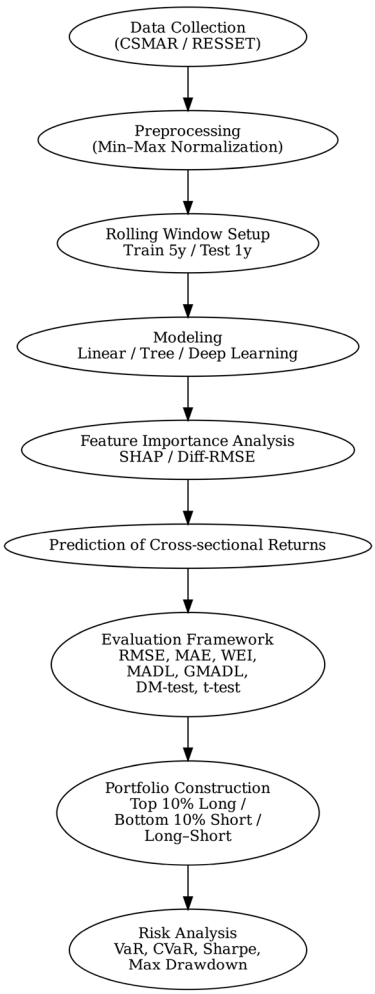


Fig. 1 Methodological framework

Feature explainer

In this study, we use Shapley's value to evaluate the contribution of each feature to the model prediction, which is derived from cooperative game theory and proposed by Shapley (1953), aiming at assigning a fair and reasonable attributional weight to each feature by calculating the marginal contribution of each feature under different combinations of features. Its mathematical definition is based on a weighted average of all possible feature permutations, ensuring that the assignment of feature importance satisfies the key properties of efficiency, symmetry, dummy property, and additivity. With these properties, the Shapley value is widely used in machine learning for model interpretation to quantify the impact of each input variable on the prediction results, thus enhancing the interpretability and transparency of the model. The formula for calculating the Shapley value is:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [\nu(S \cup \{i\}) - \nu(S)] \tag{1}$$

ϕ_i The Shapley value for feature i, representing the contribution of the feature to the model's prediction.

N: The set of all features.

S: A subset of features excluding $i, S \subseteq N \backslash \{i\}$.

$v(S)$: The value function for the feature subset S, typically measuring the performance (e.g., prediction accuracy) when using only the features in S.

$|S|!$: The factorial of the number of elements in subset S, representing the number of possible permutations of subset S.

$|N|!$: The factorial of the total number of features N, used for normalization over all possible permutations.

Shapley values are widely used to interpret machine learning models, particularly black-box models such as random forests and neural networks, by quantifying the contribution of individual features to model predictions. They are also applied in reinforcement learning and recommender systems to assess the impact of specific behaviors or recommendations. A key advantage of Shapley values is their fairness and transparency in evaluating feature importance. However, their computation becomes increasingly complex and resource intensive as the number of features grows. To mitigate this challenge, optimization techniques such as Monte Carlo sampling and approximation algorithms are commonly employed to enhance computational efficiency.

Predictor

Lasso (L1 regularization)

Lasso (Least Absolute Shrinkage and Selection Operator) is a regression method that employs L1 regularization to enforce sparsity by shrinking some coefficients to zero, facilitating feature selection. It is particularly effective for high-dimensional data but is sensitive to the regularization parameter λ and may exhibit instability with correlated features. Despite these limitations, Lasso remains widely applied in factor selection and dimensionality reduction, especially in financial modeling, where it helps identify the most relevant factors. The objective function of Lasso is:

$$\hat{\beta} = \arg\min_{\beta} \left(\sum_{i=1}^{N} \left(y_i - x_i^{\mathrm{T}} \beta \right)^2 + \lambda \sum_{j=1}^{p} \left| \beta_j \right| \right) \tag{2}$$

The L1 regularization term $\lambda \sum_{j=1}^{p} \left| \beta_j \right|$ encourages some of the regression coefficients to be pushed to zero. This not only shrinks the coefficients but also performs feature selection, removing irrelevant features. Therefore, Lasso is a technique that both builds a regression model and selects important features. Unlike L2 regularization (ridge regression), Lasso uses the L1 norm, which leads to sparse solutions, i.e., some regression coefficients are exactly zero, effectively reducing the number of features used in the model. L2 regularization only shrinks coefficients without setting them to zero.

Time-series neural network

Recurrent neural networks (RNNs) have been widely applied in financial time series modeling, including asset returns, yield curves, and macroeconomic indicators (Kijewski et al. 2024; Saad et al. 1998; Tino et al. 2001). However, standard RNNs often struggle to capture extended temporal patterns due to gradient vanishing or explosion, limiting their effectiveness in scenarios where market responses unfold gradually or with delay (Bengio et al. 1994). To address this, LSTM networks introduce memory cells and gating mechanisms to retain economically meaningful signals over longer horizons (Hochreiter & Schmidhuber 1997). This allows LSTMs to capture delayed earnings reactions, policy lags, and prolonged volatility regimes. However, their complexity leads to longer training times and higher computational costs, posing challenges for large-scale financial applications. The gated recurrent unit (GRU), introduced by Chung et al. (2014), simplifies the LSTM architecture by using fewer gates while retaining similar predictive performance. It effectively captures asset return persistence, autocorrelations, and short- to medium-term market dynamics, making it well suited for financial forecasting (Fig. 2).

RNNs are a type of neural network designed to handle sequential data by maintaining a hidden state h_t that stores historical information and updates at each time step:

$$h_t = \tanh\left(W_h x_t + U_h h_{t-1} + b_h\right) \tag{3}$$

where x_t is the input vector, W_h, U_h, b_h are learnable parameters, and $\tanh(\cdot)$ is the nonlinear activation function.

A key limitation of RNNs is the vanishing gradient problem, which hampers learning long-term dependencies. LSTM addresses this by introducing a memory cell and three gates (forget, input, and output) to control information flow. GRU simplifies this structure by removing the memory cell and using only two gates (update and reset) to manage information (Fig. 3).

Mathematical Formulation of GRU:

(1) Update Gate:

$$z_t = \sigma\left(W_z x_t + U_z h_{t-1} + b_z\right) \tag{4}$$

The update gate controls how much of the previous hidden state h_{t-1} should be retained.

(2) Reset Gate:

$$r_t = \sigma\left(W_r x_t + U_r h_{t-1} + b_r\right) \tag{5}$$

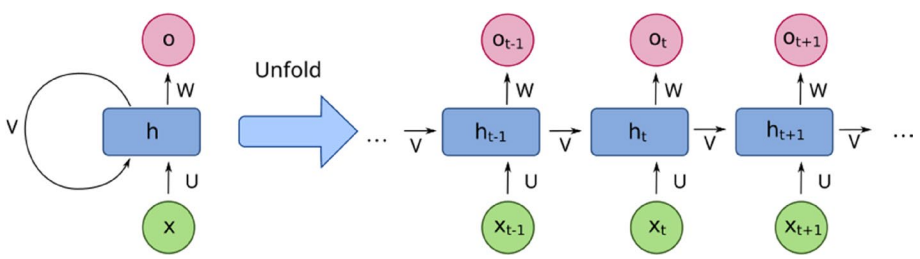


Fig. 2 Recurrent neural network architecture

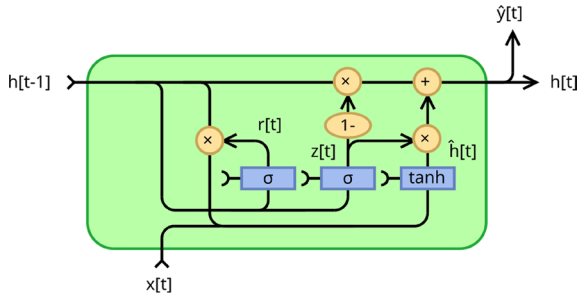


Fig. 3 Internal structure of a GRU unit

The reset gate determines how much of the past hidden state h_{t-1} should be forgotten.
(3) Candidate Hidden State:

$$\widetilde{h_t} = \tanh\left(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h\right) \tag{6}$$

(4) Final Hidden State:

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \widetilde{h_t} \tag{7}$$

If $z_t \approx 1$, the model retains the previous hidden state h_{t-1}.
If $z_t \approx 0$, the model relies more on the new candidate state $\widetilde{h_t}$.

GRU offers a simpler structure and higher computational efficiency than LSTM, using only two gates (update and reset) with fewer parameters and faster training. It converges quickly, is less prone to overfitting, and effectively captures long-term dependencies with a reduced risk of gradient vanishing. Its lightweight design makes it well suited for large-scale data, real-time tasks, and resource-constrained environments such as mobile or embedded systems.

Transformer (attention mechanism)

The Transformer architecture, originally developed for natural language processing, has recently gained traction in financial applications due to its ability to model long-term dependencies and complex interactions in sequential data. Unlike traditional recurrent or convolutional networks, the transformer processes entire sequences in parallel and assigns dynamic attention weights to different time points, making it particularly effective for high-frequency or multiperiod financial time series where patterns span extended horizons (Fig. 4). In this study, we adapt the Transformer for time-series forecasting (Fig. 5), using its multihead self-attention mechanism to identify and weigh critical signals—such as earnings announcements, volatility spikes, or macroeconomic shocks—based on their relevance to future asset returns. The feedforward layers further capture nonlinear transformations of these features, enabling the model to learn rich, context-aware representations and enhance predictive accuracy in financial forecasting.

To implement rolling cross-sectional return prediction, the Transformer is estimated within a multiyear rolling-window framework, where for each forecast period, the model is trained on the preceding historical sample and then used to generate predicted scores $\hat{y}_{i,t+1}$ for all stocks in t, which are subsequently ranked to form long–short portfolios. For each stock i at time t, the input consists of a standardized characteristic

vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$, where the full set of contemporaneous characteristics is treated as a single token. Stacking these vectors across stocks yields an input tensor $\mathbf{X} \in \mathbb{R}^{B\times L\times F}$, with B denoting the cross-sectional batch size, L the sequence dimension implied by the data structure, and F the feature dimension. Each token is mapped into the model space using an embedding of dimension $d_{\text{model}} = F$, allowing the numerical characteristics to serve directly as the model's representation. Given the nature of the input sequence, no positional encoding is needed. The resulting representations are processed by Transformer encoder layers, summarized through a pooling operation, and passed to a regression head to produce the forecast $\hat{y}_{i,t+1}$. This formulation explicitly specifies the input representation, token definition, tensor notation, and embedding structure while enabling the transformer to capture nonlinear relationships among characteristics for cross-sectional return prediction.

The Transformer model consists of an encoder and a decoder, both built from repeated layers that include multihead self-attention, FFN, residual connections, and layer normalization. At the core of the transformer is the self-attention mechanism, which captures dependencies between elements in a sequence by computing similarity scores. Given an input sequence $X \in \mathbb{R}^{n\times d}$, where n is the sequence length and d is the feature dimension of each element. the model first maps X is linearly mapped to three sets of vectors (Query, Key, and Value):

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$W_Q, W_K, W_V \in \mathbb{R}^{(d\times d_k)}$ is the trainable weight matrix, and $d_k = d/h$ is the dimension per attention head.

The attention scores are computed using the scaled dot product:

$$\text{Attention}(Q, K, V) = \text{Soft max}\left(\frac{QK^{\text{T}}}{\sqrt{d_k}}\right)V \tag{8}$$

Finally, the outputs of all attention heads are concatenated and passed through a linear layer:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W_O \tag{9}$$

with each head computed as $\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$.

The output is then processed with a residual connection and layer normalization:

$$Z = \text{LayerNorm}(X + \text{MultiHead}(Q, K, V)) \tag{10}$$

Through the above steps, the self-attention mechanism of the transformer dynamically captures global dependencies from the input sequence and generates efficient contextual representations. This mechanism provides powerful tools for time series modeling and feature extraction, making it especially suitable for predicting and analyzing high-dimensional financial data.

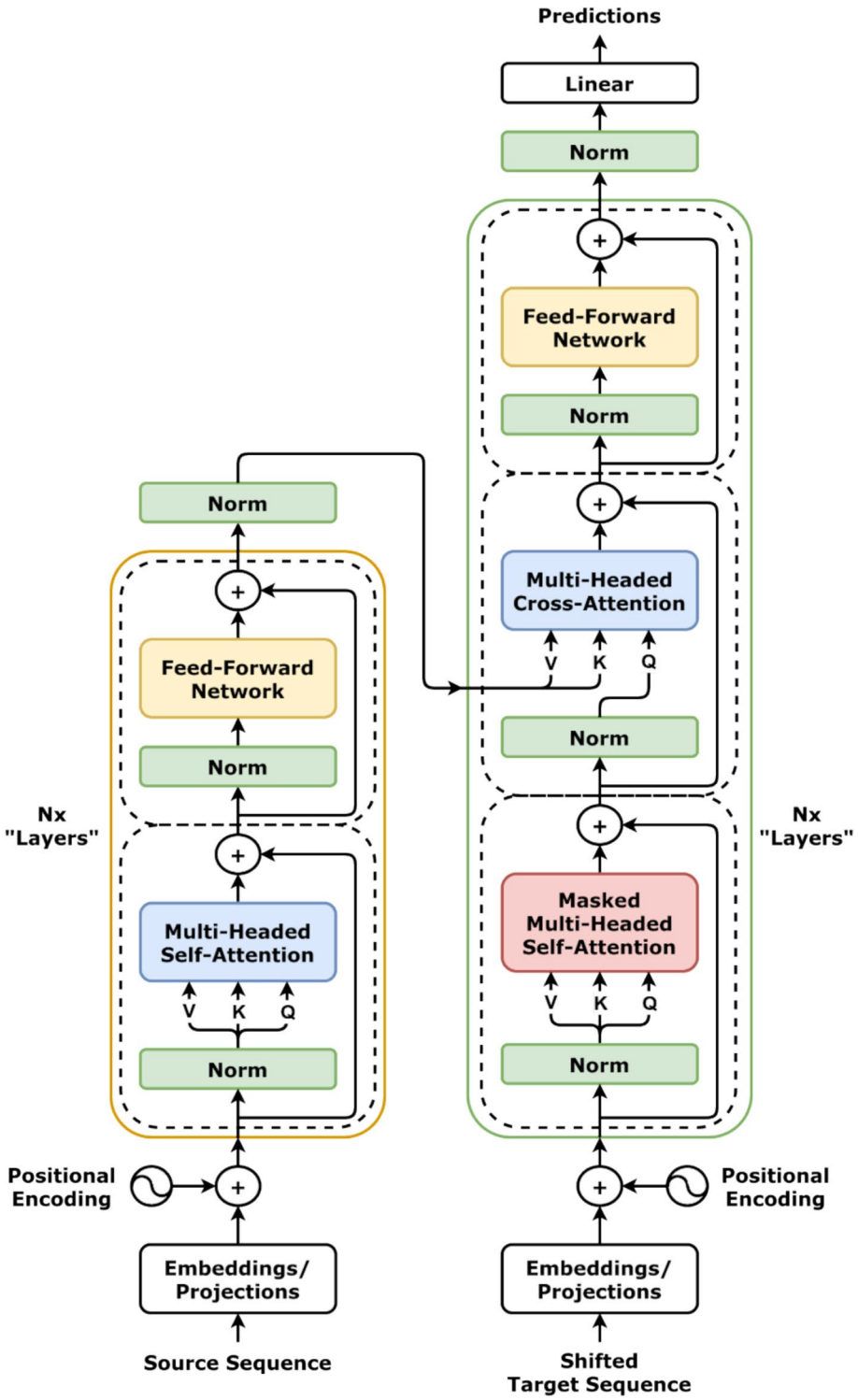


Fig. 4 Transformer network structure (Vaswani et al. 2017)

Empirical study

In this study, deep learning and machine learning models will be used to calculate the Diff-RMSE metrics using rolling training to mine the characteristic factors affecting the

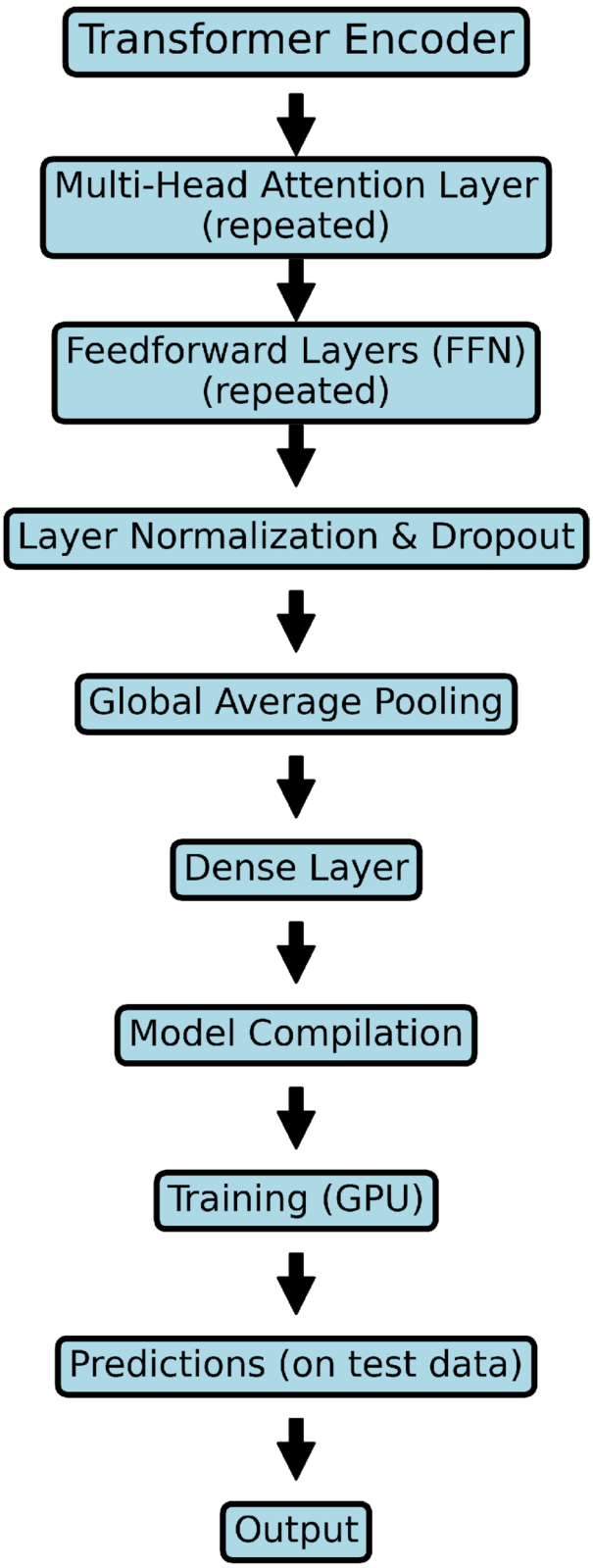


Fig. 5 Transformer-based model for enhanced prediction

performance of stocks and compare the performance of different models. First, factor importance is extracted using the L1 regularization model (Lasso), machine learning (bagging and boosting), and deep learning methods to check factors' positive or negative impact on the next period's return in different models from a cross-sectional perspective. Second, based on the performance of different models, long and short stock portfolios are constructed to compare the robust effect of the models, design long and short strategies, and comprehensively evaluate the strategies by risk-adjusted indicators such as return, Sharpe ratio, and maximum retracement. Finally, the performance of different models in terms of strategy return, risk control, and stability is compared to assess the effectiveness of various models in stock portfolio strategies (Fig. 6).

The stock portfolios are constructed by rolling training, using a rolling window of 6 years, in which the first 5 years of data are used as the training set and the last year of data as the test set. The training process for each rolling window uses the historical data of the first 5 years to train the deep learning and machine learning models to predict next-period returns in the coming year, and the performance of the models is evaluated with the test set data. This approach ensures the model's adaptability across time and avoids overfitting the data while improving the model's generalization ability and stability. The test set for each rolling window serves as an independent validation sample to simulate unseen data in the actual market.

We select hyperparameters using a rolling-origin, expanding-window cross-validation protocol with three folds. At each of the three folds, the model is trained on all data up to time t and evaluated on the next block of unseen observations; we then average the validation errors across folds to identify the hyperparameter setting that minimizes this mean error. Finally, the model is refit on the entire dataset using the chosen hyperparameters. This procedure preserves temporal ordering while yielding a robust estimate of out-of-sample performance. Detailed hyperparameter search ranges and configurations are provided in Table B.2 (Appendix B).

On this basis, we will construct stock portfolios based on the model's predictions, with the top 10% of stocks with the highest predicted returns as portfolios and the bottom 10% of stocks with the lowest predicted returns as portfolios. In this way, we can

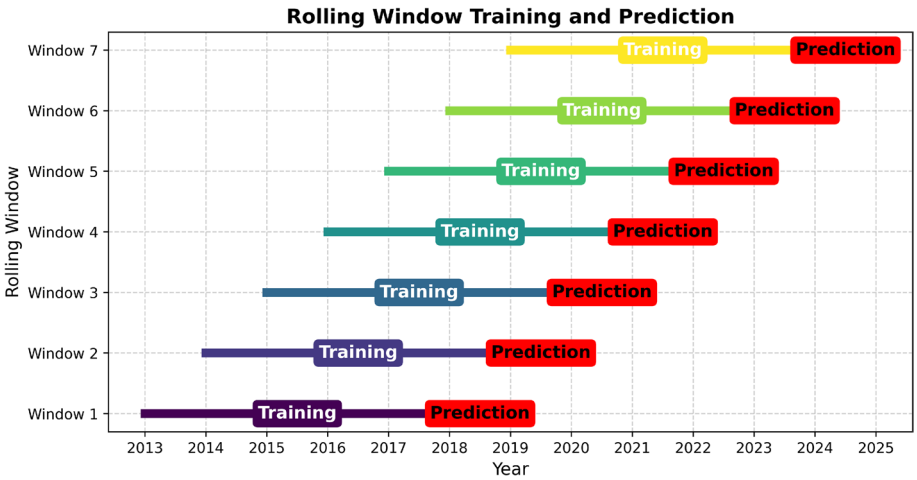


Fig. 6 Rolling window framework for model training and forecasting

compare the performance of different models in portfolio construction and evaluate the effectiveness of each model in the long-term strategy by comprehensively evaluating risk-adjusted indicators such as return, Sharpe ratio, and maximum retracement. The rolling training approach not only simulates the investment process in the real market but also effectively avoids model overfitting, ensures adaptability in different market cycles, provides more reliable strategy evaluation, and provides strong support for actual investment decisions. Combined with the back-testing results of different models, we can analyze the models' performance in different market environments more in depth, helping investors to identify which models can maintain higher returns and lower risks in complex financial markets. At the same time, this rolling training strategy also helps assess the stability of various models to ensure that they have a certain degree of risk resistance and robustness in uncertain and volatile market environments to provide a scientific basis for final portfolio optimization and strategy development.

Data

The data analyzed for the current study include the market data, company financials, and macroeconomic indices related to the A-share stocks from 2013 to 2024. To ensure the quality and reliability of the data, ST companies were excluded from the analysis because these companies usually face risk factors such as financial problems and poor governance structure of listed companies, which may interfere with the results of the study to a greater extent. We selected 30 feature factors (Table 1) and applied min–max normalization across the cross-sectional data to ensure consistency and comparability across different companies. This normalization process transformed the data into a range of [0, 1], eliminating the influence of varying magnitudes and scales among different features. Macroeconomic factors were not included in this analysis primarily because the model is designed for cross-sectional prediction at the individual stock level, where firm-specific characteristics are more directly relevant. In addition, macroeconomic indicators typically have lower frequency and may not provide sufficient granularity to capture the short-term variations in stock-level returns. Incorporating macro factors could introduce noise or temporal misalignment without significantly improving predictive power in a high-frequency cross-sectional context.

The selected factors in Table 1 cover the main categories emphasized in the asset pricing and empirical finance literature—market, liquidity, valuation, profitability, technical and risk characteristics—with a focus on their relevance for the A-share market. Liquidity-related variables capture microstructure effects in a market where retail investors and short-term trading are dominant; Liu et al. (2019) show that size and value premia in China are closely tied to liquidity conditions. Valuation ratios (PE, PB, PCF, PS) reflect mispricing relative to fundamentals in the spirit of Fama–French-type models, while profitability indicators are motivated by investment-based pricing frameworks (Hou et al. 2015). Short- and long-horizon momentum and reversal are represented by MOM3, MOM12, RSI3, RSI12 and OBV, which are particularly important in speculative markets with limited short-selling opportunities. Finally, Volatility, MomentumVol and VolChange summarize risk and uncertainty at the stock level. For each month, we first remove ST firms and stocks with missing key variables. We then apply cross-sectional min–max normalization to each factor, rescaling its values to the [0,1] interval. This

Table 1 Factors affecting model performance

Factor	Category	Description
ClPr	Market	Closing price at the end of a trading session
Mrettmv	Market	Market return weighted by tradable market value
Monrfret	Market	Return on a risk-free asset as a benchmark
Trdvol	Liquidity	Total number of shares traded in a period
Trdsum	Liquidity	Total monetary value of traded shares
MonTrdTurnR	Liquidity	Trading volume relative to outstanding shares
AvgDtrdTurnR	Liquidity	Daily average trading volume relative to outstanding shares
FundFlow	Liquidity	Total capital flow of the stock
TurnoverRatio	Liquidity	Measure of how frequently securities are traded
PE	Valuation	Price-to-earnings ratio, measuring stock price relative to earnings per share
PB	Valuation	Price-to-book ratio, assessing stock valuation against book value
PCF	Valuation	Price-to-cash flow ratio, evaluating firm value relative to cash flow
PS	Valuation	Price-to-sales ratio, indicating stock valuation relative to revenue
EPS	Profitability	Earnings per share, reflecting company profitability per share
ROE	Profitability	Return on equity, measuring profitability relative to shareholders' equity
AccumFundPS	Profitability	Accumulated reserves and surplus funds per share
OpPrfPS	Profitability	Operating profit allocated per share
NAPS	Profitability	Net assets divided by outstanding shares
IncomePS	Profitability	Revenue generated per share
NCFfropePS	Profitability	Net cash flow from operations per share
Volatility	Risk	Statistical measure of price fluctuations over time
MomentumVol	Risk	Momentum adjusted for volatility, assessing risk-adjusted returns
VolChange	Risk	Change in price fluctuation intensity over time
MA6	Technical	6-period moving average, smoothing price trends
MA12	Technical	12-period moving average, identifying long-term trends
MOM3	Technical	3-period momentum, capturing short-term price trends
MOM12	Technical	12-period momentum, assessing longer-term price movements
RSI3	Technical	3-period relative strength index, measuring short-term price momentum
RSI12	Technical	12-period relative strength index, assessing trend strength over time
OBV	Technical	On-balance volume, indicating distribution trends

transformation harmonizes units and mitigates the influence of extreme outliers while preserving the cross-sectional ranking of firms.

Through this screening, the study focuses more on companies with sound business conditions and financial performance, avoiding bias due to negative factors specific to ST companies. Moreover, ST companies are often subject to trading restrictions, regulatory scrutiny, and abnormal return patterns, which may distort the empirical results and reduce the generalizability of the findings.

We also counted the number of listed companies each month during the analysis. We found that the number of listed companies in July each year showed a significant increase compared to other months. This phenomenon may be closely related to the specific cycle of the Chinese stock market and the seasonal fluctuations of the capital market. July is an important period after the disclosure of financial reports in the first half of the year when many companies choose to enter the capital market through IPOs. Moreover, companies may adjust their strategies and financing needs in July following

the release of financial results for the first and second quarters, prompting more companies to enter the IPO process (Fig. 7).

The empirical analysis also considers institutional features of the equity market. Our strategies rebalance using monthly signals, which implies that the T + 1 settlement rule mechanism has a limited impact on implementation ability. T + 1 restrictions affect intraday round-trip trading but do not prevent investors from adjusting positions at other horizons. Price-limit hits can delay execution for a subset of stocks on specific days, yet portfolio formation relies on closing prices over an entire horizon, reducing sensitivity to such short-lived events. Short-selling constraints are partially mitigated by interpreting the short leg of the strategy as a relative-performance overlay. Overall, these institutional characteristics do not materially alter the main conclusions about the relative performance and risk profiles of the models.

Importance of characteristics

In this section, we use two methods to find each feature's importance: Shapley and differential root mean square error (Diff-RMSE). The Diff-RMSE metrics were calculated with rolling clear factor training to assess how much each factor affects the model's prediction results and thus determine the relative importance of each factor. For a comprehensive comparison, we utilized nine different machine learning models, including traditional linear statistical models and various nonlinear models. For the specific implementation, we used window rolling training, where the training window rolls forward to simulate a real market forecasting scenario.

$$\mathrm{Diff_RMSE_j} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y_{i,-j}}\right)^2} - \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y_i}\right)^2} \tag{11}$$

$$\overline{\mathrm{Diff_RMSE}} = \frac{1}{M}\sum_{j=1}^{M}\mathrm{Diff_RMSE_j} = \frac{1}{M}\sum_{j=1}^{M}\left(\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y_{i,-j}}\right)^2} - \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y_i}\right)^2}\right) \tag{12}$$

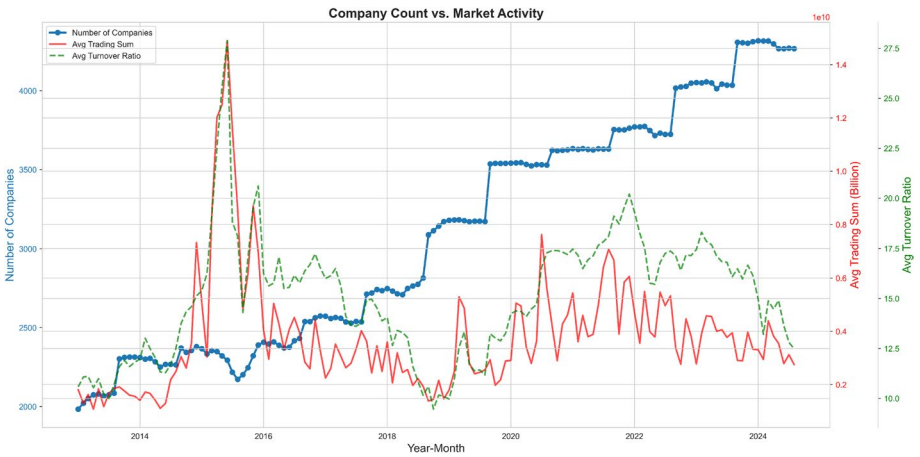


Fig. 7 Company count and market activity in the A-share market (2013–2024)

where, M: Total number of factors, N: Number of samples (rolling window size), $\widehat{y_{i,-j}}$: Prediction value after removing the j-th factor, $\widehat{y_i}$: Prediction value using all factors, Diff_RMSE_j: RMSE difference after removing the j-th factor.

We ensure the model's adaptability in market environments. To quantify the contribution of each factor in the model, the model forecasts are decomposed using Shapley values (Fig. 8), which in turn measure the impact of each factor based on the full dataset. Although SHAP values are descriptive, their patterns in Fig. 8, C.1, C.2 and C.3 correspond closely to well-established economic mechanisms. Short-horizon return signals such as Monrfret and MOM3 emerge as the most influential predictors, consistent with evidence that retail-driven markets exhibit strong reversal and transient momentum patterns due to noise trading and temporary demand shocks (De Long et al. 1990; Efficiency 1993). Liquidity-related variables and measures of trading activity, including turnover and trading volume, also rank among the most important features, in line with findings that liquidity imbalances and limited arbitrage capital generate short-run return predictability in emerging markets (Barrot et al. 2016). The SHAP interaction plots further show that valuation signals (PE, PB) contribute positively only when paired with strong profitability indicators (ROE, EPS), mirroring investment-based asset-pricing frameworks in which expected returns rise jointly with productivity and investment frictions (Hou et al. 2015). These connections demonstrate that the SHAP-based feature importance extracted from our models is economically interpretable and aligns with established theories on liquidity frictions, short-horizon behavioral dynamics, and fundamental-based pricing in the A-share market.

By calculating the Diff-RMSE metric, we can compare the performance of different models in terms of factor discovery and prediction accuracy. The SHAP-based

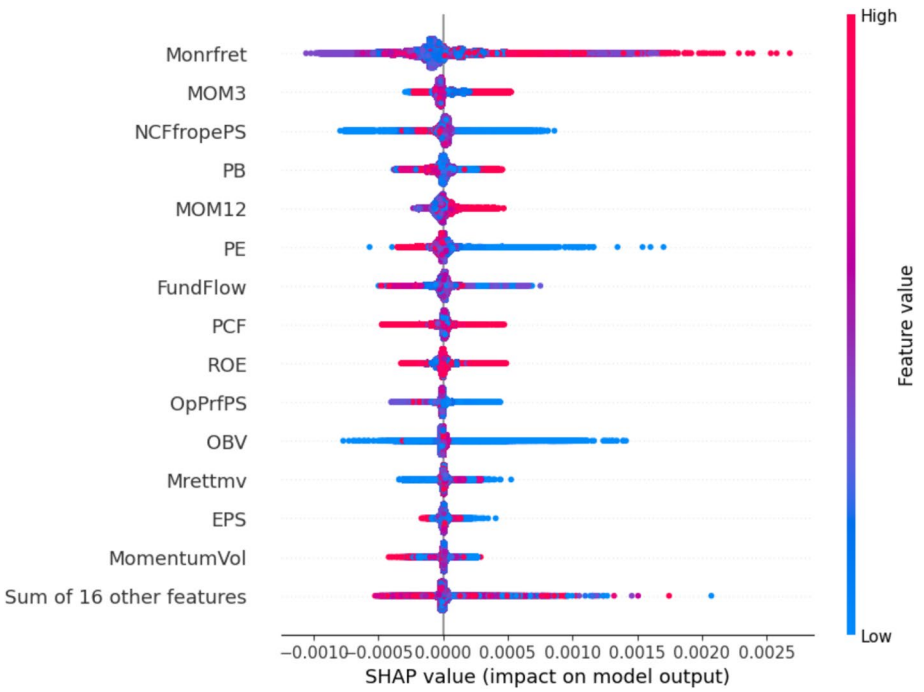


Fig. 8 Feature importance via SHAP in XGBoost

importance rankings reveal a factor hierarchy that differs markedly from the U.S. evidence in Gu et al. (2020). In the A-share market, liquidity and order-flow variables and short-horizon reversal consistently receive the highest Shapley contributions across models, whereas traditional valuation ratios and long-horizon momentum play a secondary role. This pattern is consistent with a market dominated by retail investors, where speculative trading, herding and funding constraints generate strong short-term price pressure. Profitability-related variables (ROE, EPS) remain important in deep sequential models, but their impact is more gradual, reflecting the slow diffusion of fundamental information. Together, these results highlight the need to adapt factor models to local market microstructure and institutional frictions rather than transplanting factor hierarchies from developed markets (Fig. 9).

The experimental results demonstrate that most machine learning models exhibit stronger factor discovery capabilities than traditional linear statistical methods. In particular, the Transformer and CNN-LSTM architectures—both equipped with attesntion mechanisms—show superior feature selection ability by dynamically focusing on factors that significantly influence prediction accuracy. The CNN-LSTM model, for instance, performs implicit feature selection and dimensionality reduction through convolutional layers before feeding learned representations into the LSTM for temporal modeling. These results confirm that machine learning models can effectively capture complex, nonlinear relationships among predictive signals—patterns that linear models struggle to identify—thereby offering richer support for investment factor research.

Moreover, in contrast to the U.S. stock market findings in Gu et al. (2020)—where profitability, investment, value, and momentum characteristics dominate—A-share market results highlight a different hierarchy of factor relevance. As visualized in Fig. 10, machine learning models in our setting place greater emphasis on liquidity and order-flow related variables and short-term reversal indicators, while valuation and long-term momentum signals play a comparatively secondary role. Profitability metrics such as ROE and EPS remain informative, particularly in deep sequential models with attention. These differences highlight that structural distinctions between the markets such as the dominance of retail investors trading behaviors make short-horizon and flow-based signals more predictive than in the U.S., underscoring the importance of adapting factor models to specific market environments.

Model evaluation

We use a variety of metrics to provide a comprehensive assessment of model performance, including the Diebold-Mariano (DM) test, mean square error (MSE), root mean square error (RMSE), and mean absolute error (MAE). The DM test is used to statistically compare prediction accuracies among the models, while the MSE, RMSE, and MAE assess the overall prediction error and robustness.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2 \tag{13}$$

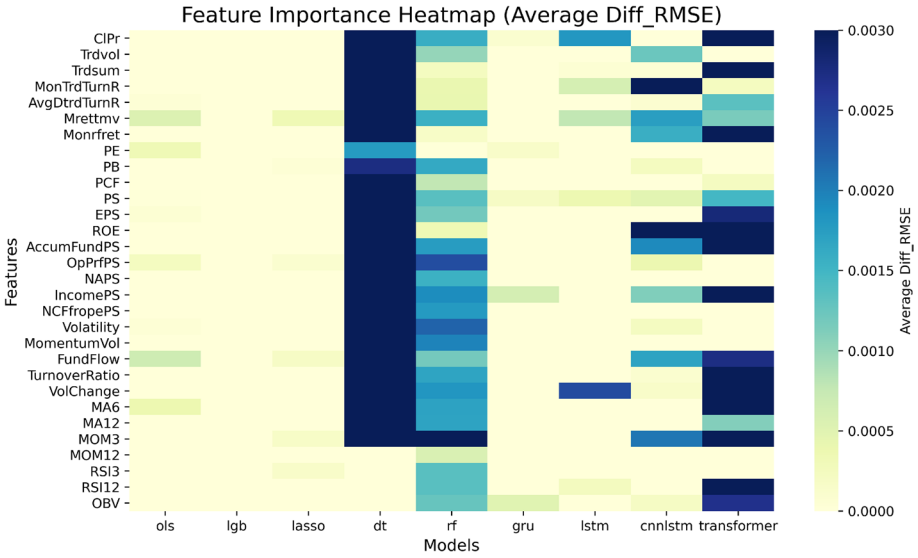


Fig. 9 Comparison of factor importance mining across models in the test set

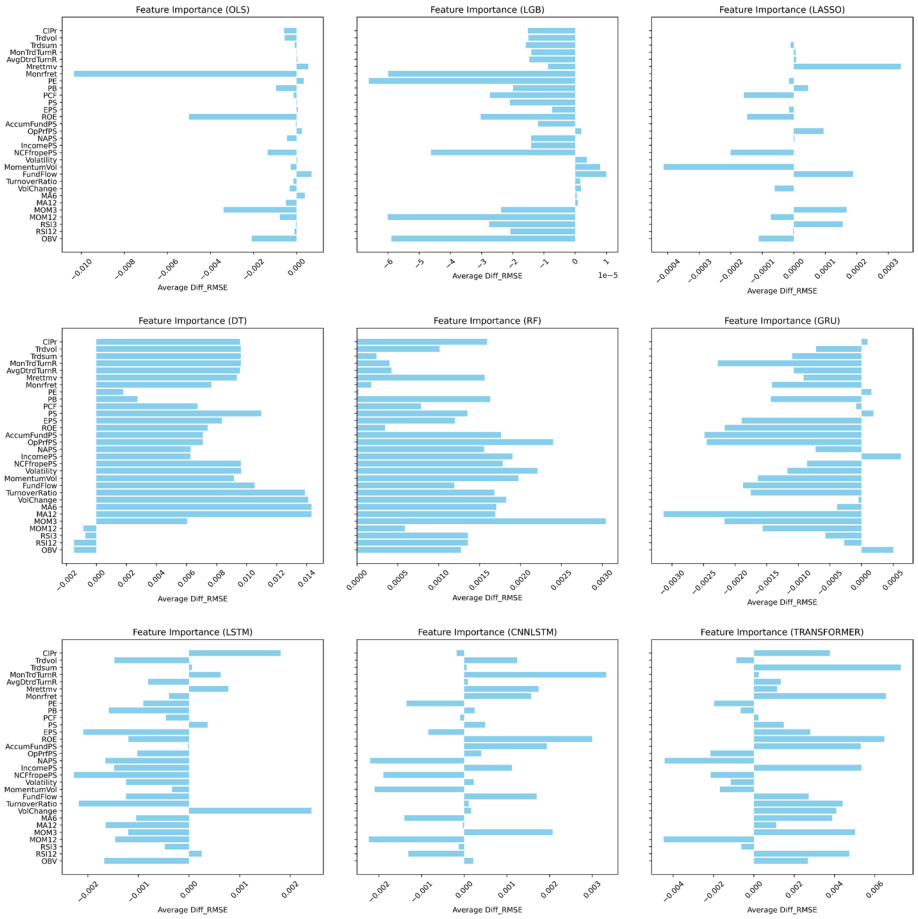


Fig. 10 Rolling window average of factor importance across different models in the test set

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y_i}\right)^2} \tag{14}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y_i}\right| \tag{15}$$

To compensate for the limitations of traditional metrics, we propose a new assessment metric, the weighted error index (WEI). Unlike MSE, which treats all samples equally and is highly sensitive to outliers, WEI incorporates weights w_+ and w_- to differentiate the importance of positive and negative samples. This flexibility allows WEI to effectively handle imbalanced datasets while reducing the influence of extreme values by relying on absolute errors instead of squared errors. WEI is a weighted loss function used to comprehensively evaluate the performance of a model's predictions. In this metric, the prediction dataset is divided into a positive sample set P and a negative sample set N, with sample sizes of n_+ and n_-, respectively.

Compared to traditional error metrics, the WEI introduces a distinct and novel design tailored to financial prediction tasks. Unlike MADL and GMADL, which emphasize directional accuracy, WEI focuses on the magnitude of prediction errors while incorporating a weighting scheme based on the relative frequency of positive and negative return days. This adjustment addresses the inherent imbalance in financial time series—where rising and falling markets may occur with unequal frequency—thereby ensuring a more robust and representative evaluation of model performance. While MADL and GMADL capture the economic cost of directional misclassification, they overlook the actual size of prediction errors. In contrast, WEI preserves the informational value of absolute errors and enhances fairness by aligning model evaluation with market structure. This makes WEI particularly well suited for regression-based financial forecasting where both predictive precision and sensitivity to asymmetric market conditions are critical.

The WEI is defined as follows:

$$WEI = w_+ \frac{1}{N_+}\sum_{t:y_t>0}\left|y_t - \widehat{y_t}\right| + w_- \frac{1}{N_-}\sum_{t:y_t<0}\left|y_t - \widehat{y_t}\right| \tag{16}$$

Here, the weighting parameters w_+ and w_- are endogenously determined based on the distribution of predicted values in the test set. Specifically, w_+ corresponds to the proportion of samples with positive predicted returns, while w_- corresponds to the proportion with negative predicted returns. y_t and $\widehat{y_t}$ denote the true and predicted values, respectively.

For a more nuanced appraisal of predictive performance, we complement WEI with two directional error metrics. The MADL penalizes only those forecasts whose sign diverges from the realized return and weights each error by $\left|y_t\right|$, thereby isolating a model's directional fidelity. Its generalized form, GMADL, further accentuates large-magnitude misclassifications by squaring each magnitude-weighted directional error, rigorously assessing robustness under extreme market swings. Collectively, WEI, MADL, and GMADL2 constitute a multi-faceted evaluation framework capturing overall accuracy, directional correctness, and resilience to severe mispredictions (Michańków et al.

2024a, 2024b). Detailed definitions and comparative analyses of these metrics are provided in Appendix A.

This metric allows flexible adjustment to emphasize either positive or negative samples, making it particularly effective for prediction tasks with imbalanced datasets. This helps mitigate the impact of imbalance on model evaluation. In this study, the WEI is calculated to compare the predictive accuracy of the model on positive and negative samples and to analyze the robustness and applicability of the model through parameter adjustments.

From the results in Table 2, LGB and LSTM perform the best in MSE and WEI metrics, achieving the lowest MSE (0.018923 and 0.019107) and WEI (0.095055 and 0.094320), respectively, which indicates that they not only have more minor overall errors but also have an advantage in evaluating the errors after considering different sample weights. In contrast, OLS performs the worst in these two metrics, with MSE and WEI of 0.025543 and 0.114781, respectively, which shows that its model is more inaccurate and less effective in error assessment after weight adjustment. MSE focuses more on the overall error, while the WEI is more in line with the differences in the sample distributions by adjusting the weights, further highlighting the robustness and superiority of the LGB and the LSTM. As seen from the DM test (Table 3), the differences in prediction errors between all models are statistically significant, indicating a significant difference in the predictive performance of the models. In specific comparisons, LGB and LSTM show superior performance, with significant t values compared to the other models, especially the poorer-performing OLS and CNN-LSTM. Meanwhile, RF and GRU perform more robustly but are slightly less healthy than LGB and LSTM. T values for OLS are positive in all comparisons with all models, indicating that its error is significantly higher than that of the other models. At the same time, CNN-LSTM and Transformer also show some disadvantages in some comparisons. Table 4 shows that most models exhibit statistically significant negative mean prediction errors, indicating a tendency to underestimate returns. In contrast, the Transformer and LSTM models have positive and statistically significant mean errors, suggesting a slight overestimation bias. Among them, the transformer yields the highest positive mean error, while linear models such as OLS suffer from the largest underestimation.

To rigorously examine the existence of systematic biases in model predictions, we perform cluster-robust t tests on the mean forecast errors, clustering at the stock level to

Table 2 Comparison of model performance on the test set

Model	MSE	RMSE	MAE	WEI	MADL	GMADL2
Ols	0.025543	0.159822	0.114442	0.114781	0.048694	0.010805
Lasso	0.019470	0.139534	0.095609	0.096073	0.046157	0.009114
DT	0.022331	0.149436	0.105390	0.105778	0.046615	0.010440
RF	0.019494	0.139622	0.094551	0.094505	0.050055	0.009954
LGB	0.018923	0.137561	0.093941	0.095055	0.045700	0.009759
GRU	0.020282	0.142416	0.097545	0.098003	0.047189	0.009909
LSTM	0.019107	0.138228	0.093804	0.094320	0.046311	0.008492
CNN-LSTM	0.020650	0.143700	0.098665	0.099090	0.051528	0.011879
Transformer	0.021586	0.146921	0.104042	0.104414	0.043605	0.008398

Table 3 Diebold-mariano test results for model comparison

	Ols	Lasso	DT	RF	LGB	GRU	LSTM	CNN-LSTM	Transformer
Ols	–								
Lasso	t = 260.98, p = 0.000	–							
DT	t = 155.93, p = 0.000	t = −84.66, p = 0.000	–						
RF	t = 180.46, p = 0.000	t = −32.70, p = 0.000	t = 126.17, p = 0.000	–					
LGB	t = 204.92, p = 0.000	t = −115.91, p = 0.000	t = 30.00, p = 0.000	t = −29.65, p = 0.000	–				
GRU	t = 229.07, p = 0.000	t = −55.59, p = 0.000	t = 57.51, p = 0.000	t = 10.99, p = 0.000	t = 76.80, p = 0.000	–			
LSTM	t = 262.94, p = 0.000	t = 149.96, p = 0.000	t = 135.56, p = 0.000	t = 86.45, p = 0.000	t = 225.72, p = 0.000	t = 198.86, p = 0.000	–		
CNN−LSTM	t = 168.38, p = 0.000	t = −185.12, p = 0.000	t = 8.55, p = 0.000	t = −40.37, p = 0.000	t = −28.32, p = 0.000	t = −114.34, p = 0.000	t = −355.02, p = 0.000	–	
Transformer	t = 194.10, p = 0.000	t = 124.63, p = 0.000	t = 132.22, p = 0.000	t = 114.67, p = 0.000	t = 139.38, p = 0.000	t = 134.88, p = 0.000	t = 99.95, p = 0.000	t = 149.14, p = 0.000	–

Table 4 Cluster-robust t-test results for mean prediction errors across models

Model	Mean Error	SE	t
Ols	−0.03685	0.000576	−63.9836
Lasso	−0.00526	0.000256	−20.5686
DT	−0.01434	0.000282	−50.9095
RF	−0.0089	0.000292	−30.4806
LGB	−0.01167	0.000226	−51.6992
GRU	−0.00752	0.000225	−33.4572
LSTM	0.000977	0.000217	4.502534
CNN−LSTM	−0.01339	0.000239	−56.0972
Transformer	0.026404	0.000552	47.79087

account for heteroskedasticity and serial correlation within firms. This approach aligns with the econometric procedures advocated by Arellano (1987) and Cameron and Miller (2015) and provides valid inference in the context of panel data where conventional i.i.d. assumptions are violated. Specifically, for each model, we regress the prediction error on a constant and compute clustered standard errors to test whether the average forecast bias significantly differs from zero. The empirical results reveal distinct patterns across model families. Penalized linear models such as LASSO exhibit statistically significant underprediction ($t = -20.57$), as do ensemble-based tree models such as LightGBM ($t = -51.70$) and random forest ($t = -30.48$), indicating a consistent downward bias in expected returns. In contrast, the transformer model produces significantly positive forecast errors (mean error = 0.0264, $t = 47.79$), suggesting systematic overestimation. The LSTM model appears nearly unbiased, although its small positive mean error remains statistically significant ($t = 4.50$). These results highlight that, beyond overall accuracy metrics, different models differ substantially in the directional reliability of their forecasts. Such evidence underscores the importance of incorporating robust hypothesis testing procedures—particularly cluster-robust inference—when evaluating predictive models in financial panel settings.

Portfolio construction

This section provides an in-depth comparative analysis of various machine learning models applied to the task of trading in financial markets, evaluating metrics such as monthly returns, cumulative returns, Sharpe ratios, and maximum drawdowns. The analysis covers long, short, and combined long-short strategies, using the CSI300 as a benchmark for performance comparison. Specifically, the long strategy targets profits by going long on the top 10% of stocks with the highest predicted returns; the short strategy analyzes market underperformance by focusing on the bottom 10% of stocks with the lowest predicted returns; and the long-short strategy integrates both approaches—simultaneously going long on the top decile and shorting the bottom decile—thereby aiming to hedge market movements and achieve more stable excess returns. This comparative framework enables a comprehensive evaluation of the practical effectiveness of different models in financial trading applications. In all strategies, we incorporate realistic trading conditions by setting a transaction cost (commission) of 0.5% and a slippage

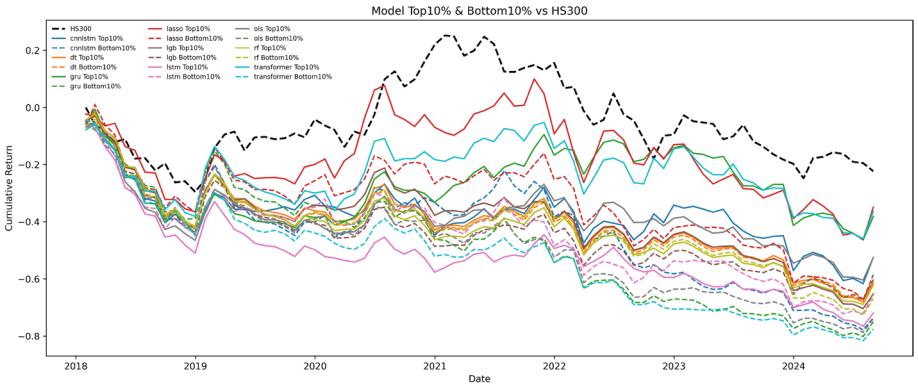


Fig. 11 Top 10% and Bottom 10% returns portfolio across different models

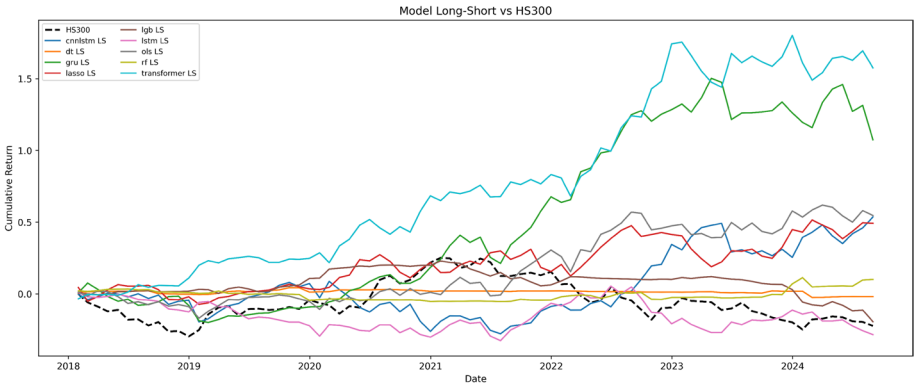


Fig. 12 Comparison of returns for long-short portfolios

rate of 0.3% to better reflect implementation feasibility in actual markets (Fig. 11 and 12).

We begin by evaluating each model's strategy using the Sharpe ratio, maximum drawdown, and return (reported in Table B.3, Appendix B). This year-by-year assessment provides a granular view of temporal stability and allows us to examine how different architectures respond to regime shifts and stress episodes, including COVID-19-induced market dislocation.

The results in Table 5 indicate substantial heterogeneity across model classes in their capacity to extract economically meaningful signals. Within the top-decile portfolios—where aggregate performance is negative due to adverse market conditions—deep learning architectures, particularly attention-based and recurrent models, experience markedly smaller drawdowns than linear and tree-based benchmarks. This relative resilience suggests that these models more effectively disentangle informative cross-sectional structure even in challenging environments. The persistent advantage of attention-based models over tree-based approaches further supports the notion that evaluation metrics incorporating directional asymmetry, such as WEI, better capture the dimensions of predictive performance that matter

Table 5 Comparative risk-return analysis of models

	CSI300	Ols	Lasso	DT	RF	GRU	LSTM	CNN−LSTM	LGB	Transformer
Cumulative Return Top10%	–	−52.46%	−34.92%	−61.42%	−62.15%	−38.16%	−71.96%	−52.44%	−65.22%	−36.36%
Cumulative Return Bottom10%	–	−73.90%	−60.23%	−60.84%	−66.85%	−75.00%	−65.58%	−73.64%	−58.55%	−77.55%
Cumulative Return Long− Short	–	54.64%	49.28%	−1.85%	10.11%	107.45%	−28.36%	53.67%	−19.34%	157.62%
Cumulative Return CSI300	−6.04%	–	–	–	–	–	–	–	–	–

for investment applications than traditional squared-error criteria. The long–short portfolios sharpen these distinctions. Deep learning models consistently generate larger and more persistent spreads between predicted winners and losers, while linear regressions and static tree-based models produce substantially weaker, and in some cases negligible, cross-sectional return differentials. This divergence indicates that models with limited representational flexibility are unable to accommodate the nonlinear interactions, state dependence, and liquidity-driven dynamics that characterize the A-share market.

We introduce two risk assessment metrics, VaR and CVaR, to comprehensively assess the risk level of different strategies. VaR measures the maximum loss a portfolio could experience over a certain period based on current positions and market volatility. Specifically, VaR gives the maximum expected loss of a portfolio at a certain confidence level (95%) over a certain period in the future (1 month). However, VaR only provides a "marginal" value of the maximum loss and does not reflect the more significant losses that can occur in extreme cases (i.e., tail risk). This is where CVaR, also known as "tail risk," plays an important role, measuring the average level of loss above the VaR threshold, which is not only concerned with the likelihood of the maximum loss but also with the severity of the loss, especially in the face of extreme market volatility (Figs. 13 and 14).

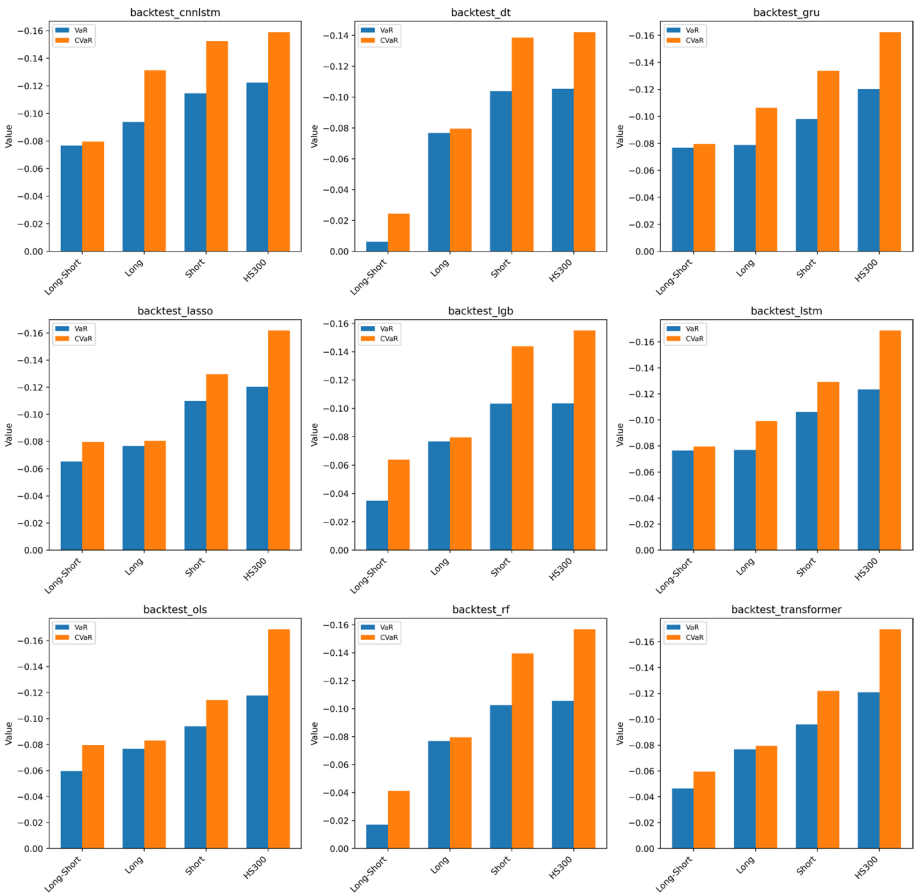


Fig. 13 Risk analysis of long-short portfolios

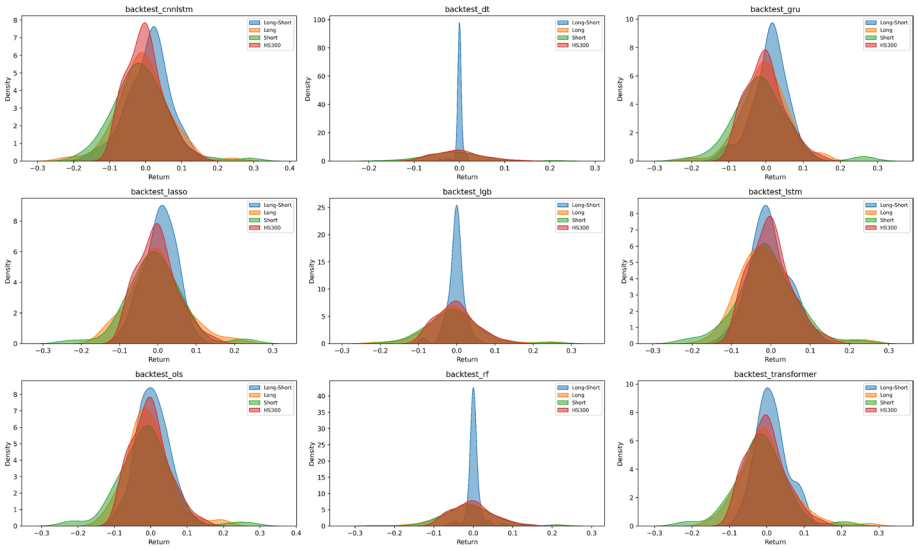


Fig. 14 Risk distribution comparison of long-short portfolios

Table 6 Risk metrics (VaR and CVaR) for long-short, top10%, and bottom10% portfolios

	Long-Short VaR (5%)	Top10% VaR (5%)	Bottom10% VaR (5%)	CSI300 VaR (5%)	Long-Short CVaR (5%)	Top10% CVaR (5%)	Bottom10% CVaR (5%)	CSI300 CVaR (5%)
OLS	-0.05949	−0.09409	−0.11768	–	−0.08307	−0.11419	−0.16868	–
Lasso	−0.06516	−0.12033	−0.10984	–	−0.08043	−0.12970	−0.16184	–
RF	−0.01696	−0.10552	−0.10251	–	−0.04117	−0.13953	−0.15679	–
DT	−0.00617	−0.10550	−0.10375	–	−0.02440	−0.14208	−0.13853	–
GRU	−0.07874	−0.09802	−0.12030	–	−0.10631	−0.13377	−0.16238	–
LSTM	−0.07631	−0.10599	−0.12353	–	−0.09914	−0.12923	−0.16871	–
CNN−LSTM	−0.09385	−0.11455	−0.12247	–	−0.13129	−0.15892	−0.15240	–
LGB	−0.03494	−0.10331	−0.10356	–	−0.06377	−0.15499	−0.14386	–
Transformer	−0.04635	−0.09600	−0.12086	–	−0.05944	−0.12187	−0.16960	–
CSI300	–	–	–	−0.07666	–	–	–	−0.07953

VaR and CVaR are risk control metrics that should not be ignored in financial markets, especially for complex products such as high-frequency trading and derivatives. By evaluating the VaR and CVaR of different trading strategies, we can help investors quantify and control risk while pursuing returns. We incorporate VaR and CVaR into the risk evaluation system to provide a comprehensive risk assessment of different strategies. This comprehensive risk management framework allows us to better understand the performance of various machine learning models in actual financial market operations, especially their ability to cope with unpredictable market risks. Furthermore, we compute these risk metrics and report them in Table B.4 (Appendix B), which enables a more granular evaluation of model performance under extreme market conditions, such as the COVID-19 shock in 2020 (Table 6).

In the analysis of VaR (5%) and CVaR (5%), the results show clear differences across models for long, short, and long–short portfolios. The pronounced gap between the Top-10% and Bottom-10% portfolios is expected and indicates that the models are

capable of meaningfully distinguishing strong from weak stocks. The Bottom-10% portfolio displays a more left-skewed and heavy-tailed loss distribution, highlighting the substantially higher risk associated with poorly ranked stocks. In contrast, the long–short portfolio exhibits a thinner-tailed distribution, and a comparison with the CSI300 index shows that the strategy substantially reduces tail risk, demonstrating strong robustness under extreme market conditions. These portfolio results also offer economically meaningful implications. The strong outperformance of Transformer and GRU in long–short returns indicates that the nonlinear patterns captured by deep architectures translate into tradable alphas rather than merely statistical improvements. The divergence between Top-10% and Bottom-10% portfolios further shows that deep learning models are particularly effective at separating winners from losers in markets dominated by retail-driven liquidity shocks and short-selling frictions. Moreover, the superior VaR and CVaR profiles of deep models are consistent with recent evidence that attention-based architectures better capture regime shifts and structural breaks.

Taken together, our empirical findings extend the literature on machine-learning-based asset pricing in several important dimensions. First, by combining Diff-RMSE and SHAP analysis, we show that nonlinear models systematically reweight factor categories toward liquidity and short-term reversal signals, thereby challenging the dominance of valuation and profitability factors emphasized in the U.S. literature. Second, the comparison of WEI, MADL, and GMADL demonstrates that deep learning architectures can simultaneously control directional errors and tail losses more effectively than linear or tree-based benchmarks. Third, the long–short portfolio evidence confirms that improvements in statistical accuracy translate into economically meaningful gains in risk-adjusted performance. Overall, these results highlight the value of finance-specific evaluation metrics and suggest that emerging-market investors can benefit from carefully designed deep learning models that explicitly account for market frictions and nonlinear factor dynamics.

Conclusion

We applied a deep learning model to study factor importance analysis and trading strategy construction on the data of the Chinese stock market for the past twelve years. The study results show that the data-driven model performs better in terms of factor importance analysis and generalization ability than traditional statistical models. To dig deeper into the model's dependence on factors, we adopt two methods to conduct a comprehensive analysis of feature importance. The first method is based on the Shapley value in game theory, which reveals the role of different factors in driving excess returns and their importance ranking by quantifying the marginal contribution of individual factors to the model's outputs. The interpretive nature of the Shapley value enables us to more intuitively understand the model's tendency to select factors under different market states. Factor selection tendencies under different market states. The second method employs Diff-RMSE analysis, which evaluates the relative importance of different factors and the impact of their combinations on the overall performance of the model by rolling out specific factors and observing the changes in the model's predictive performance. This method effectively makes up for the shortcomings of the Shapley value in the analysis of interfactor interactions, helps us further explore the optimization path of factor

combinations in trading strategies, and provides a practical reference for market investors. The study's significance lies in its validation of machine learning's transformative power in financial markets and its provision of invaluable insights for the next generation of quantitative investment strategies. By leveraging advanced machine learning techniques, such as the multihead attention network based on the transformer model, the study pushes the boundaries of factor importance selection, driving more accurate and insightful predictions. This comprehensive approach ensures that financial models are not only more accurate but also more robust and adaptable to changing market conditions.

Regarding model evaluation, we have not only adopted traditional evaluation indicators, such as return level, maximum retracement, and Sharpe ratio but also constructed WEI indicators. Compared with traditional model evaluation indicators, the WEI is more applicable to the financial market environment and can more accurately measure the model's performance. The indicator considers the model's ability to adapt to different market conditions, thus providing more informative evaluation results. With the introduction of WEI, we will be able to more comprehensively compare the strengths and weaknesses of different models and optimize our trading strategies in the future, improving the reliability and robustness of real trading applications. Regarding strategy evaluation, our analysis focuses on model selection and comprehensively considers the practical application value of the strategies from the perspective of risk-return characteristics. By constructing a two-level divergence portfolio based on factor importance analysis, we verify the stability and robustness of the model in the market environment. In addition, the designed long-short trading strategy optimizes risk control with dynamic position management and evaluates risk-return indicators such as return level, maximum retracement, Sharpe ratio, etc. The performance of the time series neural network (GRU and LSTM) and attention neural network (Transformer) is excellent in the strategy, and it has successfully achieved a return of more than 100%.

Current deep learning-based trading strategies still face many challenges in real-world applications, one of which is that dynamic changes in market prices may lead to slippage in trade execution. Since deep learning models are mainly trained based on historical data and factors such as market liquidity, order book dynamics, and trading shocks are often challenging to model accurately, the execution price may deviate from the theoretical price predicted by the model in the real market environment, thus affecting the actual return of the strategy. In addition, although the window factor neutralization and Shapley methods are used in this paper to test the factors' importance to ensure the robustness of factor selection, the nonlinear interaction effect of factor combinations remains a complex issue. The combination of different factors may have unintended effects on returns and even lead to unstable model performance under different market states. Future research could also explore the influence of macroeconomic variables at different frequencies (e.g., weekly vs. quarterly) on model stability and forecasting power. Therefore, how to systematically analyze the synergies among factors and their impact on trading returns is a direction worth exploring in depth.

Appendix

Appendix A Definitions of evaluation metrics

Weighted error index (WEI)

WEI combines absolute errors on positive and negative return days with weights equal to their sample frequencies.

$$WEI = w_+ \frac{1}{N_+} \sum_{t:y_t>0} \left|y_t - \widehat{y_t}\right| + w_- \frac{1}{N_-} \sum_{t:y_t<0} \left|y_t - \widehat{y_t}\right|$$

$$N_+ = \{t | y_t > 0\}$$

$$N_- = \{t | y_t < 0\}$$

$$w_+ = \frac{N_+}{N_+ + N_-}, w_- = \frac{N_-}{N_+ + N_-}$$

y_t: The actual return on day t, representing the observed value from the market or asset.

$\widehat{y_t}$: The predicted return on day t, generated by a forecasting model or algorithm.

N_+: The set of all positive-return days $y_t > 0$. The cardinality of this set (i.e., the number of such days) is denoted as $|N_+|$, representing how often the market exhibits upward movements during the evaluation period.

N_-: The set of all negative-return days $y_t < 0$. The size $|N_-|$ reflects the frequency of downward market movements in the sample.

w_+: The weight assigned to positive-return days, calculated as the proportion of such days relative to the total number of days with nonzero returns.

w_-: The weight assigned to negative-return days, defined analogously.

This allows for a balanced contribution of errors on both upward and downward days, even when the sample is imbalanced (e.g., more up days than down days).

Mean absolute directional loss (MADL)

MADL penalizes only those forecasts whose sign disagrees with the true sign, weighted by $\left|y_t\right|$.

$$MADL = \frac{1}{T} \sum_{t=1}^{T} \left|y_t\right| 1_{\{\text{sign}(\widehat{y_t})\text{sign}(y_t)<1\}}$$

Generalized MADL (GMADL)

GMADL applies an exponent $\gamma > 1$ to emphasize large-magnitude directional errors.

$$GMADL_\gamma = \frac{1}{T} \sum_{t=1}^{T} \left(\left|y_t\right| 1_{\{\text{sign}(\widehat{y_t})\text{sign}(y_t)<1\}}\right)^\gamma$$

In particular, setting $\gamma = 2$ (denoted GMADL2) applies a quadratic penalty to large sign-errors, thereby stressing robustness in extreme market swings.

Appendix B

See Tables B.1, B.2, B.3, B.4, B.5,

Table B.1 Descriptive statistics of standardized financial characteristics

Feature	Min	25%	50% (Median)	75%	Max	Mean	Std	Skewness	Kurtosis
ClPr	0	0.0047	0.0108	0.0331	1	0.0278	0.0459	6.3669	87.8020
Trdvol	0	0.0055	0.0135	0.0315	1	0.0290	0.0537	7.2176	84.3273
Trdsum	0	0.0061	0.0146	0.0345	1	0.0328	0.0607	6.4648	65.0109
MonTrdTurnR	0	0.0267	0.0519	0.1028	1	0.0825	0.0925	3.1328	15.3670
AvgDtrdTurnR	0	0.0261	0.0509	0.1012	1	0.0815	0.0921	3.1528	15.5237
Mrettmv	0	0.1460	0.4587	0.8865	1	0.4913	0.3687	0.0933	−1.4013
PE	0	0.3420	0.4623	0.5953	1	0.4716	0.1654	0.1228	−0.6064
PB	0	0.1538	0.3235	0.5789	1	0.4055	0.3000	0.6025	−0.8747
PCF	0	0.1880	0.4530	0.6195	1	0.4242	0.2160	−0.0415	−1.4417
PS	0	0.0006	0.0028	0.0045	1	0.0040	0.0218	37.4799	1619.4964
EPS	0	0.1144	0.1834	0.2844	1	0.2180	0.1300	1.0078	0.5855
ROE	0	0.6122	0.9224	0.9693	1	0.8109	0.2326	−1.3219	0.7236
AccumFundPS	0	0.0170	0.0362	0.0693	1	0.0500	0.0547	6.3594	77.6544
OpPrfPS	0	0.1134	0.1860	0.3278	1	0.2550	0.1963	1.3767	1.3453
NAPS	0	0.0636	0.0844	0.1171	1	0.1010	0.0621	2.8032	18.6477
IncomePS	0	0.0017	0.0038	0.0083	1	0.0092	0.0304	19.7355	539.5374
NCFfropePS	0	0.2790	0.4310	0.6732	1	0.4577	0.2392	0.2990	−0.9075
Volatility	0	0.0397	0.0724	0.1237	1	0.0954	0.0851	2.7418	13.9747
MomentumVol	0	0.1721	0.2538	0.3395	1	0.2656	0.1190	0.7553	1.1256
FundFlow	0	0.2724	0.4308	0.5935	1	0.4464	0.2075	0.1946	−0.6777
TurnoverRatio	0	0.0048	0.0110	0.0337	1	0.0281	0.0459	6.2805	86.4370
VolChange	0	0.2448	0.3266	0.4266	1	0.3458	0.1447	0.7898	0.9713
MA6	0	0.0049	0.0115	0.0350	1	0.0286	0.0455	6.1476	85.3049
MA12	0	0.0051	0.0125	0.0371	1	0.0294	0.0451	6.0180	84.5705
MOM3	0	0.3203	0.5007	0.7050	1	0.5019	0.2328	−0.0092	−1.0513
MOM12	0	0.2141	0.4549	0.6288	1	0.4405	0.2403	−0.0031	−1.1733
RSI3	0	0.2559	0.4241	0.6360	1	0.4538	0.2472	0.3335	−0.8178
RSI12	0	0.3295	0.4142	0.5136	1	0.4291	0.1464	0.5630	0.5780
OBV	0	0.1635	0.1966	0.2242	1	0.1932	0.0626	1.2080	15.5087
Monrfret_ts01	0	0.2453	0.3314	0.5380	1	0.3974	0.2356	0.8722	−0.2879

All variables are standardized to the [0,1] interval prior to summarization. Firm− level characteristics are normalized using cross-sectional min–max scaling within each period, thereby preserving the relative ranking and dispersion of firms in each cross section. The market-level return factor *Monrfret* is standardized using full-sample time-series Min–Max scaling to reflect its aggregate temporal nature rather than cross-sectional variation. Skewness and kurtosis are calculated on the standardized series and may indicate heavy-tailed behavior despite bounded support

Table B.2 Hyperparameter tuning grid employed for model training and evaluation

Model	Parameter	Range
Lasso	alpha	[0.0001, 0.0005, 0.001]
Decision Tree	max_depth	[3, 5, 7]
	min_samples_leaf	[2, 5]
Random Forest	n_estimators	[10, 20]
	max_depth	[4, 6]
	min_samples_split	[5, 10]
LightGBM	num_leaves	[5, 10]
	learning_rate	[0.01, 0.05]
	n_estimators	[100, 200]
GRU	hidden_dim	[32, 64]
	epochs	[50, 100]
LSTM	hidden_dim	[32, 64]
	epochs	[50, 100]
CNN-LSTM	hidden_dim	[32, 64]
	epochs	[10, 50]
Transformer	ff_dim	[16, 32]
	epochs	[5, 10]

Table B.3 Performance metrics of the long–short strategy for each model

Model	Year	Periods	Return	Max_Drawdown	Sharpe_Ratio
CNN-LSTM	2018	12	−0.0442	−0.0785	−0.6001
DT	2018	12	−0.0017	−0.0232	−0.9251
GRU	2018	12	−0.08	−0.1452	−0.7054
Lasso	2018	12	−0.0198	−0.0991	−0.1991
LGB	2018	12	0.0193	−0.0095	−0.0254
LSTM	2018	12	−0.125	−0.1347	−2.3189
Ols	2018	12	−0.0885	−0.1073	−1.2295
RF	2018	12	0.0096	−0.0238	−0.327
Transformer	2018	12	0.1104	−0.0176	1.0235
CNN− LSTM	2019	12	0.1223	−0.0278	0.6056
DT	2019	12	0.0145	−0.0292	−0.1286
GRU	2019	12	−0.0115	−0.0146	−0.175
Lasso	2019	12	0.0559	−0.0307	0.4801
LGB	2019	12	0.0868	−0.0321	0.9718
LSTM	2019	12	−0.1107	−0.1785	−1.1747
Ols	2019	12	0.0395	−0.0456	0.2132
RF	2019	12	−0.0542	−0.0638	−1.6185
Transformer	2019	12	0.1242	−0.0335	1.1412
CNN− LSTM	2020	12	−0.311	−0.3204	−1.4468
DT	2020	12	0.0053	−0.0122	−1.0177
GRU	2020	12	0.3036	−0.053	2.4061
Lasso	2020	12	0.1768	−0.1269	1.0187
LGB	2020	12	0.0827	−0.0072	1.1347
LSTM	2020	12	−0.1019	−0.1105	−0.5744
Ols	2020	12	0.0671	−0.0431	0.3795
RF	2020	12	−0.0076	−0.0193	−2.0393

Table B.3 (continued)

Model	Year	Periods	Return	Max_Drawdown	Sharpe_Ratio
Transformer	2020	12	0.3495	−0.0674	1.7019
CNN−LSTM	2021	12	0.2352	−0.1473	1.0881
DT	2021	12	0.0028	−0.0032	−3.8607
GRU	2021	12	0.4153	−0.1395	1.7986
Lasso	2021	12	−0.0504	−0.1182	−0.3849
LGB	2021	12	−0.1134	−0.1407	−1.9369
LSTM	2021	12	0.3377	−0.1758	1.4237
Ols	2021	12	0.2916	−0.1153	1.4996
RF	2021	12	0.0106	−0.0054	−0.5902
Transformer	2021	12	0.0878	−0.0466	0.7033
CNN−LSTM	2022	12	0.4733	−0.0447	2.232
DT	2022	12	−0.008	−0.0088	−6.9164
GRU	2022	12	0.362	−0.032	2.1972
Lasso	2022	12	0.2229	−0.0666	1.3952
LGB	2022	12	0.0356	−0.0182	0.4345
LSTM	2022	12	−0.1525	−0.2492	−0.8188
Ols	2022	12	0.1291	−0.0833	0.5955
RF	2022	12	0.0184	−0.0473	0.0076
Transformer	2022	12	0.4973	−0.0695	2.2965
CNN−LSTM	2023	12	−0.0668	−0.1589	−0.4184
DT	2023	12	0.006	−0.0095	−0.8005
GRU	2023	12	−0.0102	−0.1146	−0.1443
Lasso	2023	12	0.0241	−0.1535	0.1065
LGB	2023	12	−0.0651	−0.0811	−2.2182
LSTM	2023	12	0.1203	−0.1194	0.7032
Ols	2023	12	0.0707	−0.063	0.4157
RF	2023	12	0.0976	−0.0063	1.0428
Transformer	2023	12	0.0214	−0.1141	0.0756
CNN−LSTM	2024	8	0.3547	−0.0879	1.7956
DT	2024	8	−0.0547	−0.0428	−1.4868
GRU	2024	8	−0.1216	−0.1569	−0.6563
Lasso	2024	8	0.0461	−0.0866	0.2704
LGB	2024	8	−0.3066	−0.1473	−2.7073
LSTM	2024	8	−0.2749	−0.1805	−3.2269
Ols	2024	8	−0.0304	−0.0732	−0.4137
RF	2024	8	0.0432	−0.0579	0.2778
Transformer	2024	8	−0.1185	−0.0463	−1.0806

Table B.4 Risk metrics value at risk (VaR) and conditional value at risk (CVaR)

Model	Year	VaR Long-Short (5%)	CVaR Long-Short (5%)	VaR Long (5%)	CVaR Long (5%)	VaR Short (5%)	CVaR Short (5%)	VaR HS300 (5%)	CVaR HS300 (5%)
backtest_cnnlstm	2018	−0.0514	−0.0654	−0.1095	−0.1212	−0.1187	−0.1223	−0.0798	−0.0829
backtest_dt	2018	−0.0108	−0.0197	−0.1169	−0.1203	−0.1076	−0.1146	−0.0798	−0.0829
backtest_gru	2018	−0.0564	−0.0654	−0.1191	−0.1249	−0.1054	−0.1197	−0.0798	−0.0829
backtest_lasso	2018	−0.0794	−0.0922	−0.1050	−0.1203	−0.1064	−0.1215	−0.0798	−0.0829
backtest_lgb	2018	−0.0063	−0.0071	−0.1120	−0.1248	−0.1167	−0.1176	−0.0798	−0.0829
backtest_lstm	2018	−0.0434	−0.0561	−0.1184	−0.1186	−0.1215	−0.1340	−0.0798	−0.0829
backtest_ols	2018	−0.0410	−0.0478	−0.1135	−0.1330	−0.1081	−0.1175	−0.0798	−0.0829
backtest_rf	2018	−0.0113	−0.0167	−0.1153	−0.1176	−0.1073	−0.1138	−0.0798	−0.0829
backtest_transformer	2018	−0.0121	−0.0176	−0.0988	−0.1079	−0.1115	−0.1203	−0.0798	−0.0829
backtest_cnnlstm	2019	−0.0863	−0.1578	−0.0389	−0.0570	−0.0929	−0.1047	−0.0408	−0.0724
backtest_dt	2019	−0.0159	−0.0286	−0.0547	−0.0639	−0.0641	−0.0755	−0.0408	−0.0724
backtest_gru	2019	−0.0603	−0.1162	−0.0482	−0.0692	−0.0763	−0.1000	−0.0408	−0.0724
backtest_lasso	2019	−0.0339	−0.0540	−0.0439	−0.0656	−0.0690	−0.0752	−0.0408	−0.0724
backtest_lgb	2019	−0.0223	−0.0252	−0.0606	−0.0618	−0.0639	−0.0943	−0.0408	−0.0724
backtest_lstm	2019	−0.0458	−0.0508	−0.0757	−0.0785	−0.0539	−0.0757	−0.0408	−0.0724
backtest_ols	2019	−0.0505	−0.0885	−0.0465	−0.0468	−0.0808	−0.0921	−0.0408	−0.0724
backtest_rf	2019	−0.0275	−0.0414	−0.0618	−0.0847	−0.0628	−0.0686	−0.0408	−0.0724
backtest_transformer	2019	−0.0187	−0.0259	−0.0627	−0.0732	−0.0692	−0.0968	−0.0408	−0.0724
backtest_cnnlstm	2020	−0.1081	−0.1272	−0.0983	−0.1139	−0.1037	−0.1238	−0.0551	−0.0644
backtest_dt	2020	−0.0040	−0.0061	−0.0893	−0.0921	−0.0863	−0.0929	−0.0551	−0.0644
backtest_gru	2020	−0.0237	−0.0530	−0.0565	−0.0681	−0.0973	−0.1158	−0.0551	−0.0644
backtest_lasso	2020	−0.0477	−0.0616	−0.0877	−0.0954	−0.0906	−0.0965	−0.0551	−0.0644
backtest_lgb	2020	−0.0042	−0.0052	−0.0885	−0.0925	−0.0890	−0.0930	−0.0551	−0.0644
backtest_lstm	2020	−0.0796	−0.0925	−0.0896	−0.1046	−0.0981	−0.1224	−0.0551	−0.0644
backtest_ols	2020	−0.0498	−0.0579	−0.0732	−0.0826	−0.1044	−0.1144	−0.0551	−0.0644
backtest_rf	2020	−0.0067	−0.0089	−0.0897	−0.0930	−0.0850	−0.0934	−0.0551	−0.0644

Table B.4 (continued)

Model	Year	VaR Long-Short (5%)	CVaR Long-Short (5%)	VaR Long (5%)	CVaR Long (5%)	VaR Short (5%)	CVaR Short (5%)	VaR HS300 (5%)	CVaR HS300 (5%)
backtest_transformer	2020	−0.0452	−0.0533	−0.0921	−0.0971	−0.0853	−0.1019	−0.0551	−0.0644
backtest_cnnlstm	2021	−0.0657	−0.1063	−0.0759	−0.0967	−0.0981	−0.1337	−0.0653	−0.0790
backtest_dt	2021	−0.0012	−0.0012	−0.0639	−0.1046	−0.0662	−0.1052	−0.0653	−0.0790
backtest_gru	2021	−0.0649	−0.1006	−0.0631	−0.0799	−0.1142	−0.1439	−0.0653	−0.0790
backtest_lasso	2021	−0.0743	−0.0963	−0.0859	−0.1336	−0.0762	−0.1094	−0.0653	−0.0790
backtest_lgb	2021	−0.0347	−0.0385	−0.0698	−0.0959	−0.0551	−0.1022	−0.0653	−0.0790
backtest_lstm	2021	−0.0789	−0.1183	−0.0758	−0.0868	−0.1156	−0.1386	−0.0653	−0.0790
backtest_ols	2021	−0.0597	−0.0870	−0.0591	−0.0799	−0.1000	−0.1194	−0.0653	−0.0790
backtest_rf	2021	−0.0036	−0.0050	−0.0576	−0.1047	−0.0659	−0.1052	−0.0653	−0.0790
backtest_transformer	2021	−0.0321	−0.0466	−0.0651	−0.0959	−0.0968	−0.1327	−0.0653	−0.0790
backtest_cnnlstm	2022	−0.0428	−0.0447	−0.1253	−0.1735	−0.1401	−0.1737	−0.0781	−0.0784
backtest_dt	2022	−0.0023	−0.0026	−0.1276	−0.1587	−0.1267	−0.1584	−0.0781	−0.0784
backtest_gru	2022	−0.0274	−0.0320	−0.0829	−0.0978	−0.1662	−0.2162	−0.0781	−0.0784
backtest_lasso	2022	−0.0578	−0.0666	−0.1225	−0.1246	−0.1432	−0.1790	−0.0781	−0.0784
backtest_lgb	2022	−0.0041	−0.0048	−0.1290	−0.1623	−0.1273	−0.1606	−0.0781	−0.0784
backtest_lstm	2022	−0.0921	−0.1006	−0.0975	−0.1206	−0.1507	−0.1877	−0.0781	−0.0784
backtest_ols	2022	−0.0779	−0.0833	−0.0886	−0.0942	−0.1718	−0.2266	−0.0781	−0.0784
backtest_rf	2022	−0.0313	−0.0462	−0.1258	−0.1577	−0.1267	−0.1629	−0.0781	−0.0784
backtest_transformer	2022	−0.0385	−0.0695	−0.1119	−0.1424	−0.1699	−0.2237	−0.0781	−0.0784
backtest_cnnlstm	2023	−0.0837	−0.1339	−0.1403	−0.2210	−0.1138	−0.1783	−0.0594	−0.0621
backtest_dt	2023	−0.0043	−0.0065	−0.1129	−0.1758	−0.1136	−0.1760	−0.0594	−0.0621
backtest_gru	2023	−0.0650	−0.1048	−0.1247	−0.1966	−0.1186	−0.1641	−0.0594	−0.0621
backtest_lasso	2023	−0.0578	−0.0650	−0.0996	−0.1399	−0.1393	−0.2330	−0.0594	−0.0621
backtest_lgb	2023	−0.0224	−0.0345	−0.1328	−0.2160	−0.1152	−0.1815	−0.0594	−0.0621
backtest_lstm	2023	−0.0441	−0.0513	−0.1103	−0.1596	−0.1249	−0.2146	−0.0594	−0.0621
backtest_ols	2023	−0.0433	−0.0481	−0.0925	−0.1175	−0.1140	−0.2018	−0.0594	−0.0621

Table B.4 (continued)

Model	Year	VaR Long-Short (5%)	CVaR Long-Short (5%)	VaR Long (5%)	CVaR Long (5%)	VaR Short (5%)	CVaR Short (5%)	VaR HS300 (5%)	CVaR HS300 (5%)
backtest_rf	2023	−0.0035	−0.0059	−0.1095	−0.1699	−0.1450	−0.2453	−0.0594	−0.0621
backtest_transformer	2023	−0.0370	−0.0407	−0.0994	−0.1400	−0.1141	−0.1961	−0.0594	−0.0621
backtest_cnnlstm	2024	−0.0470	−0.0517	−0.1016	−0.1200	−0.0789	−0.0818	−0.0532	−0.0629
backtest_dt	2024	−0.0280	−0.0428	−0.0824	−0.1013	−0.0837	−0.1036	−0.0532	−0.0629
backtest_gru	2024	−0.0941	−0.1037	−0.0634	−0.0748	−0.0825	−0.0882	−0.0532	−0.0629
backtest_lasso	2024	−0.0369	−0.0439	−0.0908	−0.1166	−0.0775	−0.0801	−0.0532	−0.0629
backtest_lgb	2024	−0.0887	−0.0913	−0.0929	−0.1169	−0.0797	−0.0903	−0.0532	−0.0629
backtest_lstm	2024	−0.0650	−0.0720	−0.0865	−0.0895	−0.0780	−0.0985	−0.0532	−0.0629
backtest_ols	2024	−0.0344	−0.0378	−0.0863	−0.1120	−0.0765	−0.0778	−0.0532	−0.0629
backtest_rf	2024	−0.0382	−0.0579	−0.0778	−0.1013	−0.0925	−0.1022	−0.0532	−0.0629
backtest_transformer	2024	−0.0607	−0.0684	−0.0651	−0.0828	−0.0769	−0.0875	−0.0532	−0.0629

Table B.5 Turnover statistics of each model (2018–2024)

Model	Mean	Median	Std	Min	Max
Ols	111.44%	110.45%	18.35%	71.05%	153.95%
Lasso	102.77%	100.00%	31.30%	42.86%	180.52%
DT	178.04%	181.88%	18.94%	100.00%	200.00%
RF	170.86%	176.00%	20.47%	94.81%	190.67%
LGB	157.33%	162.75%	25.44%	62.34%	196.00%
GRU	116.40%	113.16%	25.22%	57.14%	192.11%
LSTM	112.60%	111.11%	19.62%	76.00%	186.84%
CNN-LSTM	94.24%	93.42%	30.62%	33.77%	189.61%
Transformer	102.84%	100.00%	25.58%	53.95%	185.71%

Appendix C

See Figs. C.1–C.3

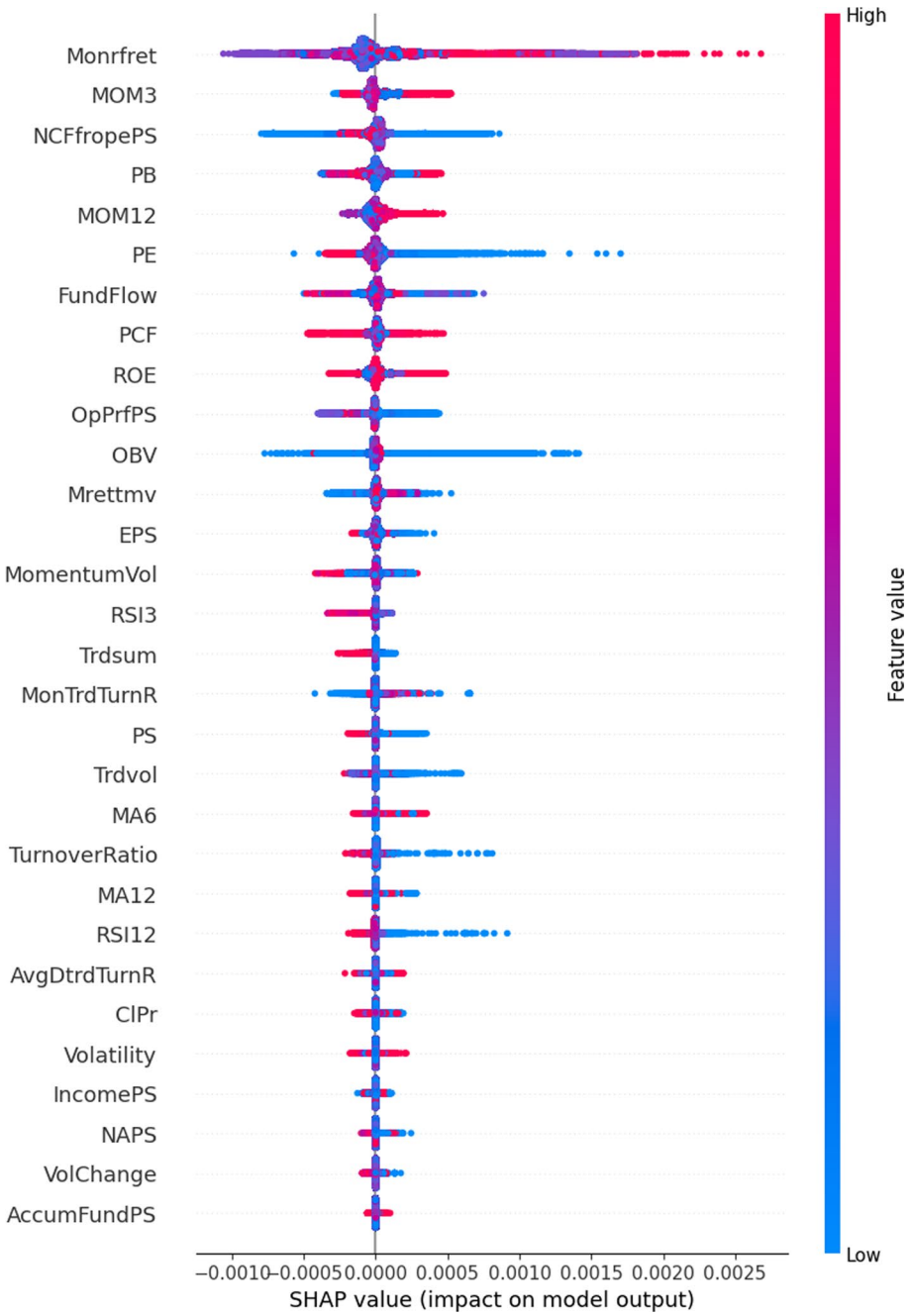


Fig. C.1 SHAP Summary Plot of All Input Features

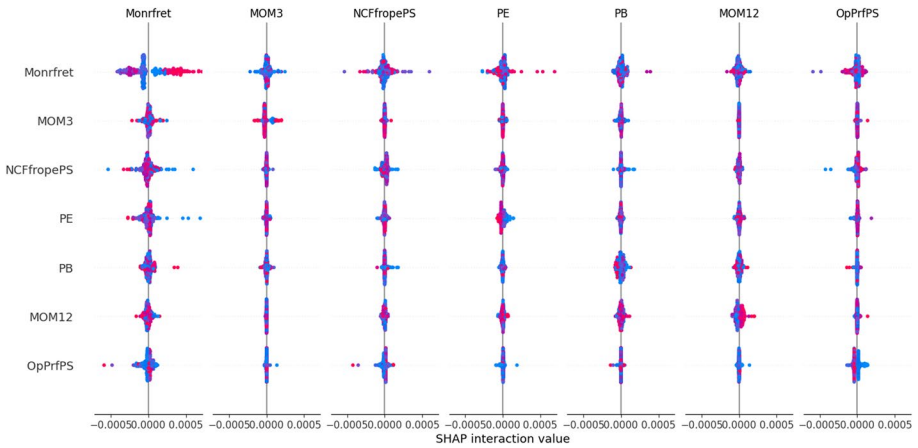


Fig. C.2 SHAP Interaction Summary of Key Financial Factors

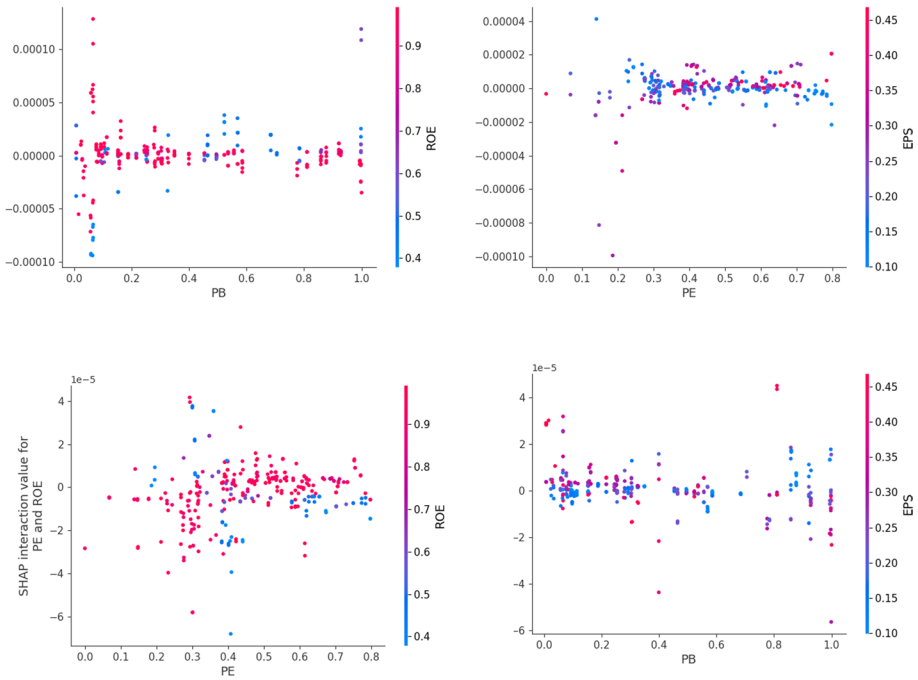


Fig. C.3 SHAP Interaction Effects Between Valuation and Profitability Factors

Figure C.1 shows the SHAP summary plot for all input features in the model. Each point represents a sample, with color indicating the feature value (red = high, blue = low). The x-axis shows how much each feature impacts the prediction. Features are ranked by importance from top to bottom.

Figure C.2 presents the SHAP interaction summary plot for the top financial features, including valuation (PE, PB), profitability (NCFfropePS, OpPrfPS), and momentum indicators (MOM3, MOM12). The density and magnitude of SHAP interaction values along

both axes highlight the presence of nonlinear dependencies among factors, addressing the factor interdependence challenge inherent in empirical asset pricing.

Figure C.3 presents SHAP dependence plots that visualize how interaction effects between valuation factors (PE, PB) and profitability indicators (ROE, EPS) contribute to model predictions. Each plot shows the SHAP interaction value between a selected feature pair, colored by a third variable, to uncover higher-order nonlinear relationships among financial predictors.

(a) PB–ROE interaction, colored by ROE
(b) PE–EPS interaction, colored by EPS
(c) PE–ROE interaction, colored by ROE
(d) PB–EPS interaction, colored by EPS

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Author contributions
HW: Conceptualization, Literature review, Methodology, Coding, Formal analysis, Investigation, Validation, Data Collection, Model evaluation, Writing—original draft, Writing—review & editing. DX: Conceptualization, Methodology, Formal analysis, Data collection, Model development, Validation, Supervision, Writing—original draft, Writing—review & editing. YD: Literature review, Data collection, Formal analysis, Writing—original draft. WX: Literature review, Data collection, Formal analysis, Writing—original draft. DC: Literature review, Data collection, Formal analysis, Writing—original draft. All authors read and approved the final manuscript.

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Data availability
The data used for the current study will be available upon request.

Declarations

Ethics approval and consent to participate
This does not apply to current research as there are no human participates involved.

Consent for publication
The authors consent for publication of the current research.

Competing interests
The authors declare that there are no competing interests for the current study.

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References

Aiolfi M, Capistrán C, Timmermann A (2010). Forecast combinations. CREATES research paper(2010–21). https://doi.org/10.2139/ssrn.1609530
Arellano M (1987) Computing robust standard errors for within-groups estimators. Oxford Bulletin of Economics & Statistics, 49(4). https://doi.org/10.1111/j.1468-0084.1987.mp49004006.x
Barrot J-N, Kaniel R, Sraer D (2016) Are retail traders compensated for providing liquidity? J Financ Econ 120(1):146–168. https://doi.org/10.1016/j.jfineco.2016.01.005
Bengio Y, Simard P, Frasconi P (1994) Learning long-term dependencies with gradient descent is difficult. IEEE Trans Neural Netw 5(2):157–166. https://doi.org/10.1109/72.279181
Bryzgalova S, Pelger M, Zhu J (2019) Forest through the trees: Building cross-sections of stock returns. Available at SSRN 3493458. https://doi.org/10.2139/ssrn.3493458
Cameron AC, Miller DL (2015) A practitioner's guide to cluster-robust inference. Journal of Human Resources 50(2):317–372. https://doi.org/10.3368/jhr.50.2.317
Carhart MM (1997) On persistence in mutual fund performance. J Financ 52(1):57–82. https://doi.org/10.1111/j.1540-6261.1997.tb03808.x
Chen L, Pelger M, Zhu J (2024) Deep learning in asset pricing. Manag Sci 70(2):714–750. https://doi.org/10.1287/mnsc.2023.4695
Christoffersen PF, Diebold FX (2006) Financial asset returns, direction-of-change forecasting, and volatility dynamics. Manage Sci 52(8):1273–1287. https://doi.org/10.1287/mnsc.1060.0520

Chun D, Cho H, Ryu D (2022) Forecasting Stock Market Volatility and Application to Volatility Timing Portfolios. *Available at SSRN* 4167561. https://doi.org/10.2139/ssrn.4167561
Chung J, Gulcehre C, Cho K, Bengio Y (2014). Empirical evaluation of gated recurrent neural networks on sequence modeling. arXiv preprint arXiv:1412.3555. https://doi.org/10.48550/arXiv.1412.3555
De Long JB, Shleifer A, Summers LH, Waldmann RJ (1990) Noise trader risk in financial markets. J Polit Econ 98(4):703–738. https://doi.org/10.1086/261703
Ding Q., Wu S, Sun H, Guo J, Guo J (2020) Hierarchical multi-scale Gaussian transformer for stock movement prediction. International Joint Conferences on Artificial Intelligence Organization, https://doi.org/10.24963/ijcai.2020/640
Efficiency SM (1993) Returns to buying winners and selling losers: Implications for. J Financ 48(1):65–91. https://doi.org/10.1111/j.1540-6261.1993.tb04702.x
Fama EF, French KR (2015) A five-factor asset pricing model. J Financ Econ 116(1):1–22. https://doi.org/10.1016/j.jfineco.2014.10.010
Feng G, Giglio S, Xiu D (2020) Taming the factor zoo: a test of new factors. J Finance 75(3):1327–1370. https://doi.org/10.1111/jofi.12883
Fischer T, Krauss C (2018) Deep learning with long short-term memory networks for financial market predictions. Eur J Oper Res 270(2):654–669. https://doi.org/10.1016/j.ejor.2017.11.054
Giglio S, Kelly B, Xiu D (2022) Factor models, machine learning, and asset pricing. Annu Rev Financ Econ 14(1):337–368. https://doi.org/10.1146/annurev-financial-101521-104735
Goulet Coulombe P, Leroux M, Stevanovic D, Surprenant S (2022) How is machine learning useful for macroeconomic forecasting? J Appl Econometrics 37(5):920–964. https://doi.org/10.1002/jae.2910
Grudniewicz J, Ślepaczuk R (2023) Application of machine learning in algorithmic investment strategies on global stock markets. Res Int Bus Finance 66:102052. https://doi.org/10.1016/j.ribaf.2023.102052
Gu S, Kelly B, Xiu D (2020) Empirical asset pricing via machine learning. Rev Financ Stud 33(5):2223–2273. https://doi.org/10.1093/rfs/hhaa009
Gu S, Kelly B, Xiu D (2021) Autoencoder asset pricing models. J Econometrics 222(1):429–450. https://doi.org/10.1016/j.jeconom.2020.07.009
Heaton JB, Polson NG, Witte JH (2017) Deep learning for finance: deep portfolios. Appl Stochastic Models Bus Ind 33(1):3–12. https://doi.org/10.1002/asmb.2209
Hochreiter S, Schmidhuber J (1997) Long short-term memory. Neural Comput 9(8):1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735
Hou K, Xue C, Zhang L (2015) Digesting anomalies: an investment approach. Rev Financ Stud 28(3):650–705. https://doi.org/10.1093/rfs/hhu068
Hou K, Xue C, Zhang L (2018) Replicating anomalies. Rev Financ Stud 33(5):2019–2133. https://doi.org/10.1093/rfs/hhy131
Hutchinson JM, Lo AW, Poggio T (1994) A nonparametric approach to pricing and hedging derivative securities via learning networks. J Finance 49(3):851–889. https://doi.org/10.2307/2329209
Jabeen A, Yasir M, Ansari Y, Yasmin S, Moon J, Rho S (2022) An empirical study of macroeconomic factors and stock returns in the context of economic uncertainty news sentiment using machine learning. Complexity 2022(1):4646733. https://doi.org/10.1155/2022/4646733
Jiang F, Lee J, Martin X, Zhou G (2019) Manager sentiment and stock returns. J Financ Econ 132(1):126–149. https://doi.org/10.1016/j.jfineco.2018.10.001
Jiang J, Kelly B, Xiu D (2023) (Re-) Imag (in) ing price trends. J Finance 78(6):3193–3249. https://doi.org/10.2139/ssrn.3756587
Kijewski M, Ślepaczuk R, Wysocki M (2024). Predicting prices of S&P 500 index using classical methods and recurrent neural networks. https://doi.org/10.62036/ISD.2024.89
Lintner J (1965) Security prices, risk, and maximal gains from diversification. J Finance 20(4):587–615. https://doi.org/10.2307/2977249
Liu J, Stambaugh RF, Yuan Y (2019) Size and value in China. J Financ Econ 134(1):48–69
Lundberg S (2017) A unified approach to interpreting model predictions. arXiv preprint arXiv:1705.07874. https://doi.org/10.48550/arXiv.1705.07874
Michańków J, Sakowski P, Ślepaczuk R (2024b) Mean absolute directional loss as a new loss function for machine learning problems in algorithmic investment strategies. J Comput Sci 81:102375
Michańków J, Sakowski P, Ślepaczuk R (2024a) Generalized Mean Absolute Directional Loss as a Solution to Overfitting and High Transaction Costs in Machine Learning Models Used in High-Frequency Algorithmic Investment Strategies. arXiv preprint arXiv:2412.18405. https://doi.org/10.48550/arXiv.2412.18405
Molnar C (2020) Interpretable machine learning. Lulu. com. https://originalstatic.aminer.cn/misc/pdf/Molnar-interpretable-machine-learning_compressed.pdf
Mullainathan S, Spiess J (2017) Machine learning: an applied econometric approach. J Econ Perspect 31(2):87–106. https://doi.org/10.1257/jep.31.2.87
Nezlobin A, Rajan MV, Reichelstein S (2016) Structural properties of the price-to-earnings and price-to-book ratios. Rev Account Stud 21:438–472. https://doi.org/10.1007/s11142-016-9356-0
Rapach DE, Strauss JK, Zhou G (2013) International stock return predictability: what is the role of the United States? J Finance 68(4):1633–1662. https://doi.org/10.1111/jofi.12041
Ribeiro MT, Singh S, Guestrin C (2016) Why should i trust you? Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, https://doi.org/10.1145/2939672.2939778
Ryś P, Ślepaczuk R (2019). Machine Learning Methods in Algorithmic Trading Strategy Optimization–Design and Time Efficiency. Central European Economic Journal, 5(52). https://content.sciendo.com/view/journals/ceej/5/52/article-p206.xml
Saad EW, Prokhorov DV, Wunsch DC (1998) Comparative study of stock trend prediction using time delay, recurrent and probabilistic neural networks. IEEE Trans Neural Netw 9(6):1456–1470. https://doi.org/10.1109/72.728395

Shapley LS (1953). 17. A Value for n-Person Games. In K. Harold William & T. Albert William (Eds.), Contributions to the Theory of Games, Volume II (pp. 307–318). Princeton University Press. https://doi.org/10.1515/9781400881970-018
Ślusarczyk D, Ślepaczuk R (2025) Optimal Markowitz portfolio using returns forecasted with time series and machine learning models. J Big Data 12(1):127. https://doi.org/10.1186/s40537-025-01164-z
Tavakoli S, Nisol G, Hallin M (2023) Factor models for high-dimensional functional time series II: estimation and forecasting. J Time Ser Anal 44(5–6):601–621. https://doi.org/10.1111/jtsa.12675
Tino P, Schittenkopf C, Dorffner G (2001) Financial volatility trading using recurrent neural networks. IEEE Trans Neural Netw 12(4):865–874. https://doi.org/10.1109/72.935096
Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need Proceedings of the 31st International Conference on Neural Information Processing Systems, Long Beach, California, USA. https://doi.org/10.5555/3295222.3295349
Wachter S, Mittelstadt B, Russell C (2017) Counterfactual explanations without opening the black box: automated decisions and the GDPR. Harv JL Tech 31:841. https://doi.org/10.2139/ssrn.3063289
Wang L, Wu Y, Huang Z, Wang Y (2024) Big data application and corporate investment decisions: evidence from A-share listed companies in China. Int Rev Financ Anal 94:103331. https://doi.org/10.1016/j.irfa.2024.103331
Welch I, Goyal A (2008) A comprehensive look at the empirical performance of equity premium prediction. Rev Financ Stud 21(4):1455–1508. https://doi.org/10.1093/rfs/hhm014
Yao J, Li Y, Tan CL (2000) Option price forecasting using neural networks. Omega 28(4):455–466. https://doi.org/10.1016/s0305-0483(99)00066-3
Zhang W, Skiena S (2010) Trading strategies to exploit blog and news sentiment. Proc Int AAAI Conf Web Soc Media. https://doi.org/10.1609/icwsm.v4i1.14075
Zhong X, Enke D (2019) Predicting the daily return direction of the stock market using hybrid machine learning algorithms. Financ Innov 5(1):1–20. https://doi.org/10.1186/s40854-019-0138-0
Zhou H, Zhang S, Peng J, Zhang S, Li J, Xiong H, Zhang W (2021) Informer: beyond efficient transformer for long sequence time-series forecasting. Proc AAAI Conf Artif Intell. https://doi.org/10.1609/aaai.v35i12.17325

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